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Project Report

AI Powered Loneliness Detection

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Abstract

Loneliness is increasingly recognized as an important social and public health concern, while the growing use of online communities provides new opportunities to investigate how individuals express loneliness through written language. This project investigates the use of Natural Language Processing (NLP), machine learning and transformer-based deep learning for automatically detecting loneliness in social media text. The study uses the FIG-Loneliness dataset, which contains Reddit posts labelled as Lonely and Non-Lonely, to develop and evaluate an explainable text-classification framework.

A quantitative experimental methodology was adopted. The dataset was explored and preprocessed before two modelling approaches were developed. Traditional machine-learning models used Term Frequency–Inverse Document Frequency (TF-IDF) representations with Logistic Regression, Random Forest and XGBoost classifiers. These models were compared with a fine-tuned DistilBERT transformer model capable of learning contextual linguistic representations. Performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, Precision–Recall curves and confusion matrices. Explainable Artificial Intelligence (XAI) techniques, particularly SHAP and LIME, were incorporated to improve model transparency and investigate influential linguistic features.

The experimental results demonstrate strong performance across all evaluated approaches. Logistic Regression was the strongest traditional model, achieving 94.50% accuracy, 94.91% F1-score and a ROC-AUC of 0.9831. DistilBERT achieved the best overall performance with 97.70% accuracy, 97.73% precision, 98.05% recall, 97.89% F1-score and a ROC-AUC of 0.9970. Explainability analysis identified interpersonal and emotion-related terms, including *friend*, *lonely*, *feel*, *talk* and *people*, among influential features within the explained model.

Overall, the findings demonstrate that transformer-based NLP can provide highly effective loneliness classification on the FIG-Loneliness dataset, while explainability methods can make predictive behavior more interpretable. The developed framework provides a useful foundation for further research into transparent AI-assisted analysis of loneliness-related online communication, while remaining a decision-support research artefact rather than a clinical diagnostic system.

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1. Problem Definition and Research Focus

1.1 Background

Social isolation has become a prominent worldwide public health issue, impacting people of various ages, cultures, and economic statuses. In contrast to transient social seclusion, loneliness is an internal emotional state defined by a perceived gap between one's ideal and real social connections. Chronic solitude has been linked to detrimental mental and physical health effects, such as depression, anxiety, cardiovascular ailments, cognitive deterioration, lowered scholastic achievement, and a decreased quality of life (World Health Organization [WHO], 2024; Surkalim et al., 2022). As online social platforms become more prevalent, people often convey their feelings, personal narratives, and mental health challenges through digital interactions, hence facilitating the early detection of loneliness by computational techniques.

Recent developments in Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML) have facilitated the automated examination of extensive textual datasets. Social media platforms like Reddit offer abundant user-generated content that mirrors emotional conditions, behavioural tendencies, and social connections. As a result, scholars have progressively investigated AI-based methods for identifying mental health disorders using linguistic examination. Notwithstanding the significant advancements made in identifying depression, anxiety, stress, and suicidal thoughts, research dedicated to the detection of loneliness remains relatively scarce, even as its societal relevance continues to escalate (Guntuku et al., 2019; Harrigian et al., 2023).

Moreover, several current predictive systems emphasise classification precision while offering no clarity into the rationale for their forecasts. In delicate areas like mental health, elucidation is crucial since healthcare practitioners and stakeholders necessitate comprehensible information prior to placing confidence in automated decision-support systems. Recent advancements in Explainable Artificial Intelligence (XAI), notably SHAP and LIME, offer frameworks for elucidating model predictions and pinpointing significant linguistic attributes, thus enhancing transparency, accountability, and user trust (Ali et al., 2023).

1.2 Problem Statement

Despite significant advancements in AI-driven mental health analytics in recent years, the precise and comprehensible detection of loneliness continues to be a rather neglected field of study. Conventional approaches to evaluating loneliness predominantly rely on self-report surveys, interviews, and psychiatric evaluations, which are laborious, resource-demanding, and frequently overlook persons who are lonely prior to the onset of severe symptoms. Numerous persons hesitate to openly share their emotional struggles, leading to postponed assistance and diminished chances for proactive help.

Social media provides a supplementary avenue for behavioural and linguistic insights that could enable the prompt recognition of loneliness. Nonetheless, current research frequently focuses on overarching mental health illnesses instead of loneliness as a separate psychological phenomenon. Furthermore, traditional machine learning models depend significantly on manually crafted textual characteristics, while transformer-based frameworks like BERT and DistilBERT have exhibited enhanced contextual language comprehension in many NLP applications. Notwithstanding these advancements, few research has methodically contrasted conventional machine learning with transformer-based models for the identification of loneliness, while concurrently integrating explainability methods to enhance model clarity. Tackling this research deficiency is crucial for creating dependable, comprehensible, and pragmatically useful AI systems that can aid in early mental health intervention.

1.3 Research Aim

The principal objective of this study is to create and assess an explainable AI framework for identifying loneliness in Reddit postings by contrasting conventional machine learning techniques with transformer-based deep learning models. The research additionally seeks to examine the language characteristics that most profoundly influence loneliness forecasting through the application of explainable AI methodologies.

1.4 Research Objectives

To achieve the research aim, the following objectives have been established:

1. To gather and prepare the FIG-Loneliness dataset for natural language processing applications.

2. To do exploratory data analysis to comprehend the attributes and distribution of textual data pertaining to loneliness.
3. To construct and assess conventional machine learning models, such as Logistic Regression, Random Forest, and XGBoost, utilising TF-IDF feature representations.
4. To construct a transformer-oriented DistilBERT model for the classification of contextual loneliness.
5. To evaluate the prediction efficacy of conventional machine learning and transformer-based models utilising Accuracy, Precision, Recall, F1-score, ROC-AUC, and Precision-Recall metrics.
6. Utilise Explainable Artificial Intelligence methodologies, such as SHAP and LIME, to discern the linguistic characteristics that affect model predictions.
7. To assess the efficacy of explainable AI in enhancing transparency and reliability in automated loneliness detection.

1.5 Research Questions

This study seeks to answer the following research questions:

- **RQ1:** Can machine learning and transformer-based deep learning models accurately detect loneliness from social media text?
- **RQ2:** Which modelling approach provides the highest predictive performance for loneliness classification?
- **RQ3:** Which linguistic features contribute most significantly to loneliness prediction?
- **RQ4:** How can explainable AI techniques improve the interpretability and transparency of automated loneliness detection systems?

1.6 Scope of the Study

This study concentrates solely on the binary categorisation of loneliness utilising textual information from the publicly accessible FIG-Loneliness dataset available on Hugging Face. The research encompasses data preprocessing, exploratory data analysis, TF-IDF feature extraction, application of traditional machine learning techniques, creation of a DistilBERT transformer model, comparative performance assessment, and interpretability analysis utilising SHAP and LIME. The initiative does not encompass the gathering of primary data, live surveillance of social media channels, clinical assessment of mental health conditions, or the implementation of a fully operational application. These limits guarantee that the study is feasible given the existing time constraints, computer capabilities, and ethical standards.

1.7 Significance of the Research

The suggested study enhances both scholarly understanding and practical implementations in AI-driven mental health analysis. From an academic standpoint, it enhances the current literature by offering a thorough comparison of conventional machine learning and transformer-based methodologies for detecting loneliness, while incorporating explainable AI into the assessment framework. From a pragmatic viewpoint, the established system could assist healthcare practitioners, educational bodies, mental health entities, and scholars by facilitating the prompt detection of individuals potentially facing loneliness via automated scrutiny of publicly accessible social media data.

The results detailed subsequently in this dissertation reveal that transformer-based models attained enhanced predictive accuracy, whereas explainability methods uncovered psychologically significant linguistic markers including friend, lonely, feel, and discuss. These results underscore the promise of integrating sophisticated NLP models with interpretable AI methodologies to develop transparent decision-support systems that might aid, rather than supplant, mental health practitioners in recognising individuals who might require prompt intervention.

2. Literature Review and Theoretical Framework

2.1 Introduction

The swift expansion of social media has revolutionised interpersonal communication, generating substantial amounts of user-created textual data that include emotions, behaviours, and psychological conditions. As a result, Artificial Intelligence (AI) and Natural Language Processing (NLP) have gained significance in diagnosing mental health disorders via automated text examination. Recent research indicates that machine learning and deep learning algorithms may proficiently identify depression, anxiety, stress, suicidal thoughts, and various psychological diseases from social media content (Harrigian et al., 2023; Chancellor & De Choudhury, 2020). Nonetheless, the identification of loneliness is still relatively unexamined, despite its acknowledged influence on both physical and mental well-being. This literature review rigorously analyses current studies on loneliness detection, machine learning, transformer-based language models, explainable artificial intelligence (XAI), and the theoretical foundations of this research. It further recognises research deficiencies that warrant the suggested comparative analysis of conventional machine learning and DistilBERT models employing explainable AI methodologies.

2.2 Loneliness Detection through Social Media Analytics

Loneliness is acknowledged as a complex psychological phenomenon that signifies felt inadequacies in social connections rather than actual social seclusion. Chronic loneliness has been linked to depression, anxiety, cardiovascular ailments, diminished cognitive abilities, and heightened mortality risk (Surkalim et al., 2022). Conventional approaches to evaluating loneliness predominantly depend on instruments like the UCLA Loneliness Scale or in-person psychological assessments. Despite their clinical reliability, these methods are laborious, costly, and inappropriate for ongoing large-scale surveillance.

The growing accessibility of publicly available social media data has allowed researchers to investigate computational methods for identifying loneliness via linguistic examination. Reddit, specifically, has emerged as a significant repository of mental health information, as members often share their personal experiences anonymously and with more depth than on numerous other sites. Research indicates that textual patterns associated with emotional expression, social connections, self-revelation, and adverse mood frequently serve as valuable indicators of psychological health (Harrigian et al., 2023).

Notwithstanding these developments, loneliness has garnered far less scholarly focus compared to depression or suicidal thoughts. Current research often regards loneliness as a subordinate variable in extensive mental health studies instead of an autonomous predictive factor. As a result, there is a restricted comprehension of which computational frameworks are best suited for loneliness categorisation and which language traits most substantially enhance prediction efficacy.

2.3 Machine Learning for Text Classification

Conventional machine learning has gained significant traction in natural language processing because to its comparatively low computational expense, clarity of interpretation, and robust prediction efficacy on organised textual formats. Prior to classification, text is typically converted into numerical vectors by feature extraction techniques like Bag-of-Words (BoW) or Term Frequency-Inverse Document Frequency (TF-IDF). TF-IDF continues to be favoured as it allocates more significance to meaningful words while diminishing the impact of frequently appearing keywords.

Numerous supervised learning methods have proven effective in text categorisation tasks. Logistic Regression continues to be a highly effective baseline classifier because to its straightforwardness, computing economy, and capacity to generalise well in high-dimensional feature spaces (Kowsari et al., 2019). Ensemble learning methods like Random Forest and XGBoost have been extensively utilised due to their ability to capture intricate nonlinear interactions and enhance predicted reliability across several decision trees (Chen & Guestrin, 2016).

Despite the fact that these methodologies have yielded commendable outcomes in numerous NLP applications, their efficacy is predominantly contingent upon manually crafted features. They typically overlook contextual connections among adjacent words and encounter difficulties with semantic ambiguity, sarcasm, and long-range interdependence commonly found in social media discourse. As a result, scholars have progressively embraced contextual language models that can acquire more nuanced semantic representations.

2.4 Transformer-Based Deep Learning Models

Transformer designs have revolutionised natural language processing by replacing sequential recurrent processing with self-attention mechanisms that understand long-range contextual links. Transformers create contextual word embeddings where the same words can have different meanings based on neighbouring content, unlike typical machine learning models.

BERT (Bidirectional Encoder Representations from Transformers) improved NLP performance by considering past and subsequent words during training (Devlin et al., 2019). DistilBERT, a simplified BERT, distils knowledge using its architecture. DistilBERT maintains over 95% of BERT's predictive accuracy while reducing model size by 40% and speeding inference (Sanh et al., 2019).

Transformer-based models outperform standard machine learning methods in sentiment analysis, emotion identification, and mental health predictions due to their superior semantic meaning capture (Harrigian et al., 2023). Their contextual representations help recognise subtle linguistic patterns that TF-IDF features cannot. Transformer models require greater processing power and are often "black-box" systems, making understanding difficult without explainability tools.

2.5 Explainable Artificial Intelligence

While prediction precision continues to be a crucial aim in machine learning, transparency has emerged as a vital consideration in critical fields like healthcare and mental health. Healthcare professionals necessitate rationales that elucidate the reasoning behind a model's specific prediction prior to placing their trust in automated decision-support systems.

Explainable Artificial Intelligence (XAI) seeks to enhance clarity by elucidating the elements that affect model forecasts. Prominent among the prevalent explainability techniques are SHAP (SHAPely Additive explanations) and LIME (Local Interpretable Model-Agnostic Explanations). SHAP quantifies the impact of certain variables on total model predictions through the lens of cooperative game theory, facilitating both global and local insights into model behaviour (Lundberg & Lee, 2017). Conversely, LIME elucidates singular predictions by developing locally interpretable surrogate models centred on particular observations (Ribeiro et al., 2016).

Recent studies in healthcare indicate that explainability methods enhance trust, accountability, fairness, and adherence to regulations by allowing users to comprehend the rationale behind certain forecasts (Ali et al., 2023). In the realm of mental health analytics, SHAP and LIME aid in recognising psychologically significant linguistic indicators linked to particular emotional states. Thus, explainability becomes an essential element of ethical AI development rather than simply an ancillary visualisation method.

2.6 Theoretical Framework

This study is grounded in three synergistic theoretical frameworks: Natural Language Processing theory, supervised machine learning theory, and Explainable Artificial Intelligence.

The notion of Natural Language Processing posits that written language encompasses quantifiable linguistic patterns that can reflect fundamental psychological conditions. By means of tokenisation, preprocessing, vectorisation, and contextual embedding, significant semantic data can be derived from textual communication and converted into formats comprehensible by machines.

Supervised learning theory offers the secondary theoretical basis. Supervised classification systems acquire prediction associations by correlating labelled training instances with established output categories. In this research, the loneliness classifications from the FIG-Loneliness dataset allow the algorithms to differentiate between lonely and non-lonely messages by recognising distinctive textual features.

The third conceptual basis is Explainable Artificial Intelligence. XAI enhances predictive modelling by augmenting interpretability, allowing researchers and practitioners to comprehend both overall feature significance and specific prediction dynamics. This theoretical viewpoint corresponds with contemporary suggestions for transparent AI systems in healthcare, where elucidation fosters responsible decision-making and enhances stakeholder confidence. Collectively, these three theoretical viewpoints offer the conceptual foundation for the suggested comparative assessment of conventional machine learning, transformer-driven deep learning, and explainability methodologies.

2.7 Research Gap

Existing literature reveals many serious shortcomings that justify our inquiry. Much research has focused on depression, anxiety, and suicide risk prediction, but little on loneliness identification as a categorisation issue. Thus, evidence on the best computer approaches for predicting loneliness is rare. Second, several research compare conventional machine learning models or transformer-based frameworks without using the same datasets or evaluation criteria. This makes it difficult to determine if computing complexity always improves forecast accuracy. Third, many published studies highlight anticipated precision but lack model projection logic. Lack of explainability reduces receptivity and hinders practical adoption in healthcare settings where understanding forecasts is key. Few research use contextual transformer models and Explainable Artificial Intelligence to identify the most important linguistic variables for loneliness prediction. This presents a substantial research opportunity as transparent AI systems become essential for ethical healthcare deployments.

2.8 Chapter Summary

The literature suggests that AI can assess mental health using social media information. Traditional machine learning models provide good baseline results, while transformer-based frameworks improve language contextual comprehension. Still, loneliness detection is understudied, and few studies use predictive modelling and explainable AI to clarify. This research addresses these limits by carefully analysing Logistic Regression, Random Forest, XGBoost, and DistilBERT models on the FIG-Loneliness dataset using SHAP and LIME for predictive interpretation. This unified framework provides the theoretical and methodological foundation for the research design in the next chapter.

3. Research Design and Methodology

3.1 Research Design

This research employs a quantitative experimental methodology to examine the efficacy of Artificial Intelligence (AI) methods for identifying loneliness from social media text. A quantitative methodology was used as the main aim is to assess, contrast, and analyse the predicted efficacy of various classification models by objective statistical measures. Empirical investigation facilitates the methodical juxtaposition of algorithms under uniform circumstances, permitting performance discrepancies to be ascribed to the learning techniques instead of fluctuations in the dataset or assessment procedure (Creswell & Creswell, 2023).

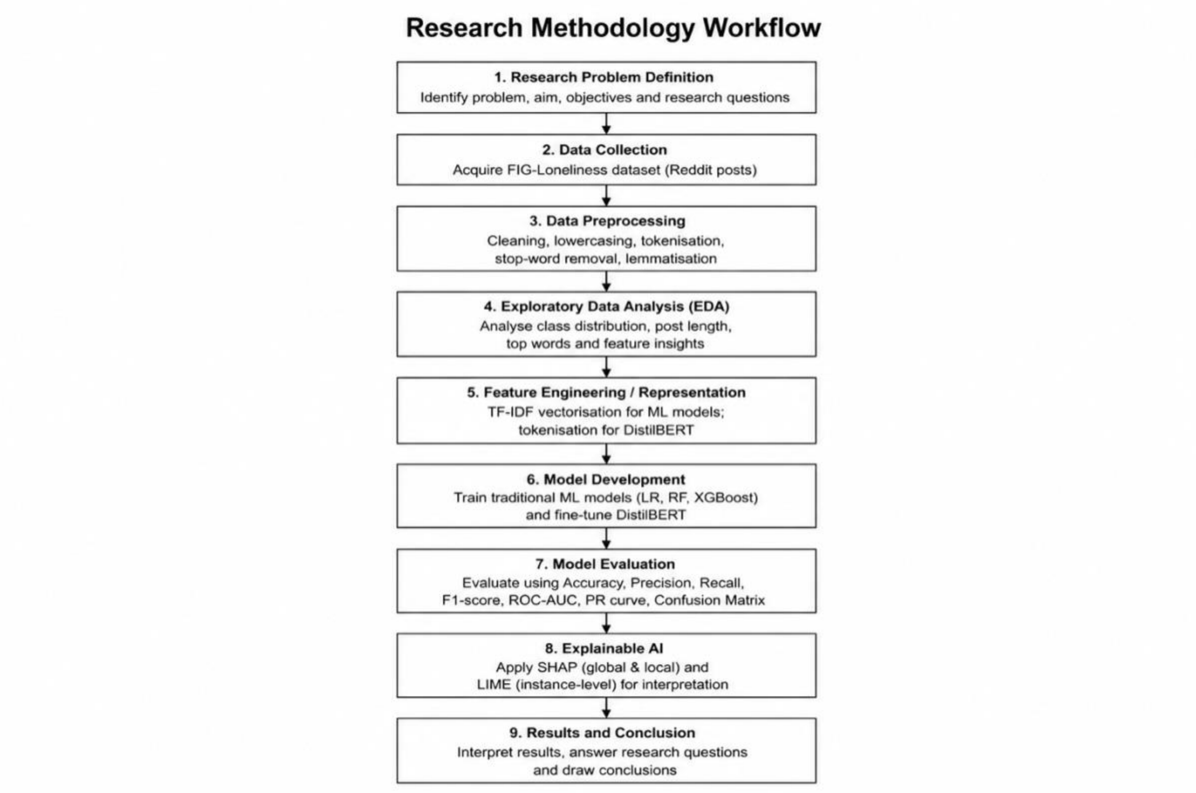

Figure 1 ***Figure 3.1. Research Methodology Workflow for Explainable Loneliness Detection***

The research methodology depicted in Figure 3.1 adheres to a systematic experimental framework aimed at exploring automated loneliness identification using social media text. The procedure commences with the delineation of the study issue, purpose, aims, and enquiries, succeeded by the procurement and analysis of the FIG-Loneliness dataset, which comprises Reddit posts categorised as Lonely or Non-Lonely. Data preprocessing is then executed to enhance textual quality by cleaning, tokenisation, stop-word elimination, and lemmatisation. Exploratory Data Analysis (EDA) is subsequently employed to examine class distribution, characteristics of post length, and significant vocabulary trends within the dataset. Subsequent to data preparation, two synergistic approaches for text representation and modelling are examined. Conventional machine-learning methods employ TF-IDF representations alongside Logistic Regression, Random Forest, and XGBoost, while the transformer-based methodology refines DistilBERT to acquire contextual representations directly from tokenised text. All models undergo uniform assessment utilising accuracy, precision, recall, F1-score, ROC-AUC, Precision-Recall analysis, and confusion matrices. This comparison method allows for a systematic evaluation of the relative strengths of traditional and transformer-based NLP techniques. Ultimately, Explainable Artificial Intelligence methodologies are integrated to enhance the clarity of the classification procedure. SHAP is employed to examine both global and local feature contributions, whereas LIME facilitates interpretation at the instance level. The workflow thus amalgamates data analysis, predictive modelling, comparative assessment, and explainability into a cohesive research framework, allowing for the simultaneous consideration of predictive efficacy and model clarity in tackling the research enquiries.

The study contrasts three conventional machine learning algorithms Logistic Regression, Random Forest, and XGBoost with a transformer-based deep learning architecture, DistilBERT. The research additionally integrates Explainable Artificial Intelligence (XAI) methodologies to elucidate model behaviour and enhance transparency. This experimental design explicitly facilitates the research objective of pinpointing the most precise and comprehensible method for detecting loneliness.

3.2 Data Source and Collection

The study employs the FIG-Loneliness dataset accessible via the Hugging Face repository. The collection comprises publicly accessible Reddit posts categorised as Lonely and Non-Lonely, rendering it suitable for supervised binary text classification. Reddit was chosen due to its users' propensity to anonymously share personal experiences, feelings, and psychological struggles, yielding valuable textual data for mental health examination.

Given that the material is publicly accessible and entirely anonymised, there was no necessity for primary data gathering involving human subjects. This greatly diminished ethical hazards while facilitating reproducible experiments. The dataset features a fairly equitable class distribution, reducing bias from significant class imbalance and facilitating dependable model assessment.

3.3 Data Preprocessing

Data preprocessing improved text quality and prepared the dataset for machine learning and transformer-based modelling. This preprocessing sequence eliminated duplication, managed absence values, converted text to lowercase, removed punctuation and special symbols, tokenised, removed stop words, and lemmatised. These methods reduced textual clutter while preserving meaning.

Traditional machine learning models used TF-IDF to convert sanitised text into numerical feature vectors. TF-IDF is ideal for high-dimensional text categorisation since it emphasises significant words and downplays keywords (Kowsari et al., 2019). DistilBERT used its own sub-word tokeniser to create contextual embeddings from source text, eliminating feature engineering.

3.4 Model Development

The research assesses conventional machine learning techniques alongside transformer-based deep learning architectures to deliver a thorough comparison of various Natural Language Processing methodologies.

Three supervised machine learning algorithms were implemented using the Scikit-learn library:

- **Logistic Regression**, selected because of its simplicity, computational efficiency, and strong performance in high-dimensional text classification.
- **Random Forest**, chosen for its ensemble learning capability, which improves prediction robustness through multiple decision trees.
- **XGBoost**, included because of its gradient boosting framework that has consistently achieved state-of-the-art performance across many classification problems.

To explore contextual language comprehension, DistilBERT was utilised using the Hugging Face Transformers framework. DistilBERT is a streamlined transformer framework originating from BERT through the process of knowledge distillation. It maintains the majority of BERT's forecasting proficiency while considerably diminishing computing demands, rendering it appropriate for research settings with constrained hardware assets (Sanh et al., 2019).

This amalgamation of classical and transformer methodologies facilitates an impartial evaluation of feature-engineered learning against contextual language modelling.

3.5 Model Evaluation

The dataset was partitioned into training and testing subsets to assess generalisation efficacy on novel data. The evaluation of model performance utilised various complementing criteria, guaranteeing a thorough assessment of categorisation quality beyond just overall accuracy.

The evaluation metrics included:

- Accuracy
- Precision
- Recall
- F1-score
- Receiver Operating Characteristic Area Under the Curve (ROC-AUC)
- Precision–Recall (PR) Curve
- Confusion Matrix

Accuracy evaluates forecast validity, whereas Precision measures the percentage of lonely posts correctly detected. Recall tests models' ability to identify actual loneliness, which is crucial in mental health applications where missed detections can have major consequences. The F1-score evaluates Precision and Recall comprehensively, while ROC-AUC and Precision-Recall curves examine discriminatory efficacy across categorisation thresholds.

All machine learning models performed well in the testing, but DistilBERT had the highest accuracy and fewest classification errors, demonstrating its contextual understanding of Reddit posts.

3.6 Explainable Artificial Intelligence

To strengthen the clarity of the model, Techniques of Explainable Artificial Intelligence were included into the assessment procedure. Two distinct approaches to explainability were utilised:

- SHAP (SHapley Additive exPlanations)
- LIME (Local Interpretable Model-Agnostic Explanations)

SHAP was employed to deliver both overarching and specific elucidations by assessing the impact of distinct textual attributes on predictive results. The resultant feature importance graphs highlighted significant terms like friend, lonely, feel, and talk, indicating that the model acquired linguistically relevant markers linked to loneliness.

LIME enhanced SHAP by elucidating individual forecasts via locally interpretable surrogate models. This facilitated an in-depth analysis of the impact of particular words on personal classification choices. The integration of SHAP and LIME improves transparency, bolsters stakeholder trust, and facilitates the ethical implementation of AI in mental health contexts (Ali et al., 2023).

3.7 Software Tools and Technologies

The initiative was executed wholly in Python, utilising Google Colab as the development platform. Data preprocessing and manipulation were executed utilising Pandas and NumPy, whereas Scikit-learn offered implementations of conventional machine learning methods and assessment metrics. DistilBERT was created utilising the Hugging Face Transformers framework, employing PyTorch as the foundational deep learning library.

Data visualisation was executed via Matplotlib and Seaborn to produce exploratory data analysis graphics, confusion matrices, ROC curves, Precision-Recall curves, and feature significance diagrams. The SHAP and LIME libraries were utilised to produce comprehensible elucidations of model forecasts.

The utilisation of open-source software improved reproducibility, decreased implementation expenses, and fostered transparent scientific inquiry.

3.8 Methodological Strengths and Limitations

The chosen method has many benefits. The experimental approach first allows unbiased comparison of machine learning paradigms using the same datasets and assessment metrics. Second, Explainable AI addresses the growing need for transparent and reliable AI solutions in healthcare. Thirdly, a publicly available benchmark dataset improves reproducibility and allows comparison with subsequent studies.

Several limits must be considered. The study only used Reddit data, which may not accurately reflect terms used on other social media sites or in clinical settings. The information is limited to multilingual demographics because it only includes English posts. Transformer models require more computer resources than typical machine learning methods, but DistilBERT improves forecast accuracy. Since loneliness is a complex psychological experience impacted by social and environmental factors, the developed models should be used as decision-support tools rather than clinical diagnostic systems. Despite these limits, the technique provides a strong, unambiguous, and replicable structure for analysing AI-driven loneliness identification and meets study aims.

4. Artefact Design and Development

4.1 Overview of the Developed Artefact

The main artefact produced in this study is a loneliness detection system empowered by Explainable Artificial Intelligence (XAI) that autonomously categorises Reddit posts as Lonely or Non-Lonely through Natural Language Processing (NLP), conventional machine learning methods, transformer-based deep learning, and explainability strategies. The artefact was created to examine if contextual language models surpass traditional machine learning methods while also enhancing model transparency via Explainable Artificial Intelligence.

In contrast to traditional text classification systems that prioritise only predicted accuracy, the created artefact incorporates explainability inside the classification process. This facilitates a comprehensive understanding of model activity on a global scale while providing localised explanations for specific predictions, rendering the system more suitable for delicate domains like mental health analytics. The item thus fulfils the project goals by integrating superior forecast accuracy with clear decision-making.

4.2 System Architecture

The system was designed as a modular pipeline with sequential processing steps, enabling the independent development, testing, and assessment of each component. The comprehensive workflow includes data collection, text preprocessing, exploratory data analysis, feature extraction, machine learning classification, transformer-based learning, interpretability assessment, and model evaluation.

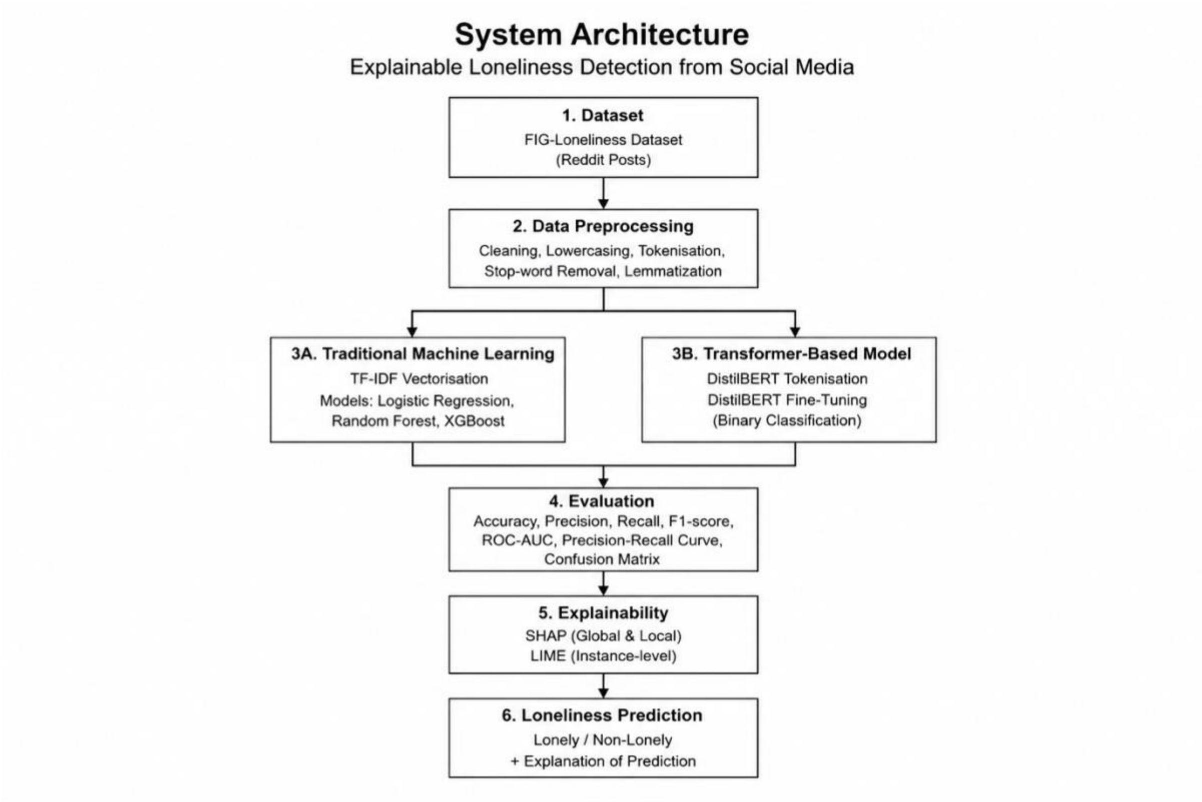

Figure 2 ***Figure 4.1. Proposed System Architecture for Explainable Loneliness Detection Using Machine Learning and Transformer-Based NLP***

The suggested framework utilises a modular pipeline where the FIG-Loneliness dataset functions as the main input source. Reddit submissions undergo preliminary data-quality assessments and text preprocessing prior to exploratory analysis aimed at investigating class distribution, post length, and vocabulary traits. The refined data are then channelled thru two concurrent modelling routes. The conventional machine-learning approach converts textual information into TF-IDF feature vectors and utilises classifiers such as Logistic Regression, Random Forest, and XGBoost. Simultaneously, the transformer pathway employs DistilBERT tokenisation and contextual embeddings to optimise a transformer-based binary classifier. This distinction facilitates a direct juxtaposition of traditional feature-engineered NLP and contextual deep-learning methodologies. Your report previously outlines this dual representation approach, employing TF-IDF for traditional models and DistilBERT producing contextual embeddings straight from tokenised text.

Both modelling approaches contribute to a shared assessment framework utilising Accuracy, Precision, Recall, F1-score, ROC-AUC, Precision-Recall analysis, and confusion matrices. These criteria facilitate uniform evaluation of model efficacy. The assessment outcomes reveal the most robust predictive model, but SHAP and LIME offer supplementary explanatory analyses. SHAP facilitates both global and local feature attribution, while LIME offers interpretation at the instance level for specific predictions. The ultimate method generates both a binary categorisation of Lonely/Non-Lonely and a comprehensible elucidation of the language attributes influencing the decision.

The process initiates by retrieving the FIG-Loneliness dataset, which is publically accessible, from the Hugging Face repository. Subsequent to preprocessing, textual information is converted into numerical formats appropriate for various learning techniques. Conventional machine learning algorithms employ TF-IDF feature vectors, but DistilBERT analyses unrefined text via contextual tokenisation and embedding creation. Ultimately, SHAP and LIME produce comprehensible elucidations that pinpoint significant linguistic attributes influencing predictive results. This modular design enhances maintainability, reproducibility, and scalability, facilitating future expansion to incorporate more datasets or classification models.

4.3 Development Environment and Technologies

The object was developed exclusively with open-source software under the Google Colab cloud platform. Google Colab was chosen due to its integrated Python functionality, GPU acceleration for training transformers, collaborative development features, and interoperability with prominent machine learning libraries.

The execution depended on certain specific Python libraries. Pandas and NumPy facilitated data manipulation and numerical analysis, whereas Scikit-learn offered implementations of Logistic Regression, Random Forest, XGBoost integration, TF-IDF vectorisation, and performance assessment measures. Deep learning was executed via the Hugging Face Transformers framework in conjunction with PyTorch, facilitating effective fine-tuning of the DistilBERT language model. Data visualisation was executed with Matplotlib and Seaborn, whereas SHAP and LIME libraries facilitated both global and local interpretability assessments.

The sole utilisation of open-source tools improves reproducibility, lowers implementation expenses, and allows subsequent researchers to duplicate the experimental setup.

4.4 Data Preprocessing and Feature Engineering

The calibre of textual data significantly impacts classification efficacy. As a result, an extensive preprocessing framework was established to eliminate noise while maintaining significant linguistic data.

Initially, redundant entries and absent values were eliminated to enhance dataset uniformity. The text was later transformed to lowercase to maintain consistency during tokenisation. Punctuation marks, number symbols, URLs, and extraneous special characters were eliminated as they provide minimal semantic value within the chosen classification models. Tokenisation divided phrases into discrete lexical components, whereas stop-word elimination removed prevalent functional terms that offer minimal distinguishing information. Lemmatisation was then utilised to condense words to their base forms while maintaining semantic significance.

Feature engineering varied based on the chosen classification algorithm. Conventional machine learning algorithms utilised Term Frequency–Inverse Document Frequency (TF-IDF) to convert sanitised text into high-dimensional numerical representations. TF-IDF highlights distinctive vocabulary while diminishing the impact of commonly occurring yet less informative terms. Conversely, DistilBERT produced contextual embeddings directly from tokenised text, facilitating the automatic capture of semantic links among words without the need for manual feature engineering.

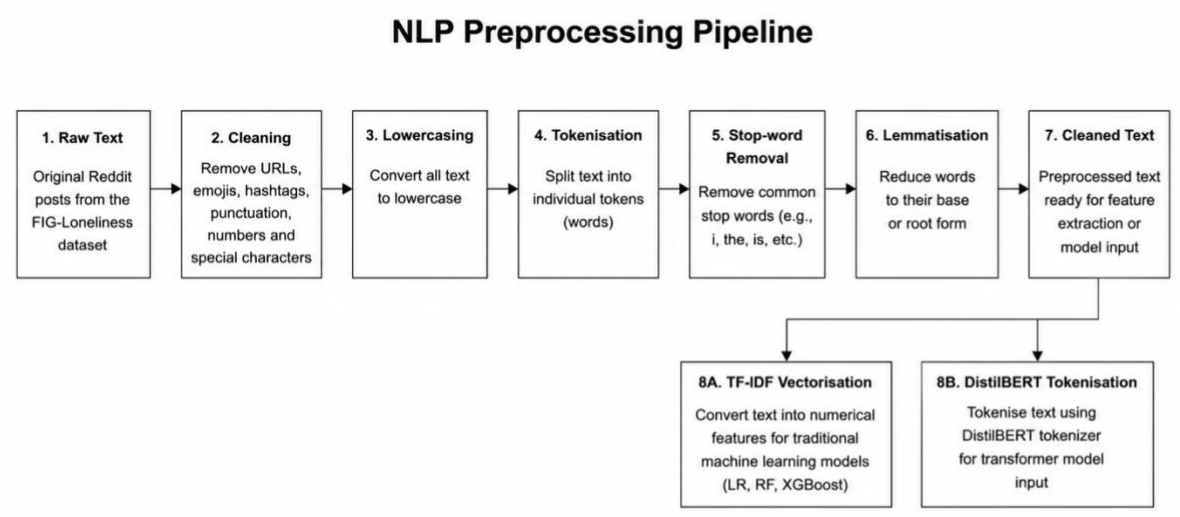

Figure 3 ***Figure 4.2. NLP Preprocessing Pipeline for the FIG-Loneliness Reddit Dataset***

Figure 4.2 depicts the chronological NLP preprocessing sequence utilised for the FIG-Loneliness Reddit submissions. The operation commences with unrefined textual data and implements cleansing techniques to eliminate extraneous components including URLs, emojis, hashtags, punctuation, numerals, and other superfluous symbols. The text is then converted to lowercase and segmented into separate lexical components. Elimination of stop-words diminishes the impact of often used terms that possess minimal distinguishing significance, whereas lemmatisation transforms inflected words into their root forms. These phases diminish textual clutter and yield a more uniform representation for future examination. Subsequent to preprocessing, the workflow diverges into two modelling trajectories. In conventional machine-learning models, the sanitised text is converted thru TF-IDF vectorisation, producing numerical feature representations appropriate for Logistic Regression, Random Forest, and XGBoost. The transformer pathway readies text for DistilBERT tokenisation and fine-tuning, allowing the model to comprehend contextual connections among words. Preserving these unique representations was essential as traditional machine-learning techniques necessitate explicit numerical feature extraction, while DistilBERT acquires contextual representations via its pretrained transformer framework. This dual feature extraction approach facilitated an equitable assessment of traditional vector-based learning vs contextual transformer representations.

4.5 Machine Learning Model Development

Three supervised machine learning algorithms assessed baseline loneliness detection performance. Logistic Regression was chosen for its computational efficiency and high-dimensional text categorisation success. Its linear decision boundary produces understandable forecasts. The bootstrap-aggregated Random Forest ensemble learning method integrated many decision trees. This method consolidates results from numerous trained trees to reduce overfitting and improve forecast consistency. XGBoost was added because gradient boosting algorithms excel at structured prediction. XGBoost improves weak learners by fixing previous trees' errors, creating exact ensemble models. All models used the same training and testing datasets, ensuring fair comparisons across evaluation measures. To reduce overfitting and improve forecast accuracy, hyperparameter adjustment was done.

4.6 Transformer-Based Deep Learning Implementation

A transformer-based model utilising DistilBERT was created to address the contextual constraints of conventional machine learning. In contrast to TF-IDF representations, DistilBERT produces contextual embeddings by analysing the interrelations of adjacent words via self-attention mechanisms. This facilitates the comprehension of intricate linguistic structures frequently encountered in Reddit dialogues. The model underwent fine-tuning utilising the annotated FIG-Loneliness dataset via supervised learning. Throughout the training process, tokenised Reddit postings were transformed into contextual embeddings prior to classification with a task-specific output layer. Fine-tuning modified the pre-trained language models for the loneliness detection task while maintaining the linguistic insights acquired during extensive pre-training. In contrast to traditional machine learning models, DistilBERT necessitated more computer power yet delivered enhanced predicted accuracy by acquiring contextual rather than frequency-oriented textual representations.

4.7 Explainable Artificial Intelligence Integration

A notable characteristic of the created artefact is the incorporation of Explainable Artificial Intelligence methodologies. Numerous high-performing AI models function as opaque systems, complicating users' comprehension of the prediction generation process. This constraint is especially concerning in healthcare contexts when clarity is crucial.

To tackle this issue, the object integrates SHAP and LIME. SHAP was utilised to ascertain overall feature significance and measure the impact of specific phrases on model predictions. The produced summary visualisations indicated that terms like friend, lonely, feel, chat, and people had the most significant impact on classification choices. These results indicate that the model acquired psychologically significant linguistic structures instead of depending on random statistical associations. LIME enhanced SHAP by elucidating specific predictions. For every chosen Reddit article, LIME discerned the terms that favourably and adversely influenced the anticipated category, facilitating a comprehensive understanding of particular categorisation results. The integration of SHAP and LIME significantly enhanced model transparency, hence facilitating the advancement of reliable AI.

4.8 System Testing and Performance Evaluation

The development methodology included extensive functional integrity and predictive capacity tests. Early exploratory data analysis confirmed the dataset's integrity, class distribution, and text properties. Every categorisation model was evaluated using uniform training and testing datasets.

Performance was assessed using Accuracy, Precision, Recall, F1-score, ROC-AUC, Precision Recall curves, and Confusion Matrices. All machine learning models had high projected accuracy, with Logistic Regression outperforming Random Forest and XGBoost. However, DistilBERT had the highest accuracy and fewest false positives and negatives. All models showed exceptional discriminative efficacy in ROC and Precision-Recall tests, with DistilBERT having the largest area under both curves.

The explainability test confirmed that SHAP and LIME's essential points aligned with psychologically meaningful loneliness language, bolstering the artifact's reliability.

4.9 Technical Contribution and Innovation

The item combines cutting-edge technologies into an explainable NLP framework to demonstrate technical proficiency. Rather than testing prediction accuracy, the system combines standard machine learning, transformer-based contextual learning, and Explainable AI into an experimental framework. The deliberate juxtaposition of feature-engineered machine learning and contextual transformer models using the same datasets, assessment criteria, and interpretability approaches improves the product. This comprehensive technique improves model clarity and impartially evaluates predicted efficacy. The prototype for loneliness detection can be developed for further research or integrated into decision-support systems for healthcare, educational, and mental health institutions. The object combines high classification accuracy with clear forecasts to improve mental health analytics AI.

5. Project Management

5.1 Project Planning and Organisation

Effective project management was necessary to complete this research on time while maintaining the quality of the object and dissertation. The project was planned, executed, assessed, and documented in a cycle. The project timeline was developed during proposal and changed during execution as new challenges and needs surfaced. This organised strategy follows project management principles like defined goals, milestone assessment, risk reduction, and continuing evaluation to achieve project success (Project Management Institute [PMI], 2021).

The effort had eight main steps: literature review, dataset investigation, data preparation, machine learning model design, transformer-based model implementation, model assessment, explainable artificial intelligence integration, and dissertation production. Segmenting the research into achievable phases allowed methodical progress tracking and ensured task interdependencies were addressed in order. After data pretreatment and exploratory data analysis, model creation could begin, but explainability investigation required trained classification models.

5.2 Timeline and Milestones

A project timetable was designed to enable time for each research stage and assure timely submission. The theoretical framework and research techniques were developed in the first weeks through literature evaluation and project planning. Data collection, preprocessing, and exploratory data analysis were used to understand FIG-Loneliness.

Logistic Regression, Random Forest, and XGBoost were implemented, followed by DistilBERT transformer model implementation and fine-tuning. Breakthroughs included comprehensive performance evaluation using many categorisation measures and SHAP and LIME integration to improve model interpretability. Final steps involved assessing experimental outcomes, visualising study findings, and writing the dissertation.

This milestone-based strategy tracked project progress and identified tasks that needed extra time or improvement. Periodic reviews demonstrated that project objectives matched research goals throughout development.

5.3 Risk Management

Throughout the project, risk management reduced completion risks. Planning identified technical and project risks. Data quality risks included missing values, duplicate records, and inconsistent text style. To reduce risk, the pretreatment process includes duplicate removal, text cleaning, tokenisation, stop-word removal, and lemmatisation. Overfitting plagued ensemble and transformer-based models. Separate training and testing datasets and performance evaluation across multiple factors rather than accuracy lowered this risk. Transformer models require far more processing power than traditional machine learning, raising computational risk. Google Colab used DistilBERT, a computationally efficient transformer architecture, for GPU acceleration and training time reduction. Finally, milestone and progress tracking eased project scheduling issues. Model optimisation and explainability analysis delays were accommodated by altering the implementation timetable while maintaining the project timeline.

5.4 Challenges and Adaptations

Project implementation presented some obstacles. Many publicly available mental health datasets focus on depression or anxiety rather than loneliness, making dataset selection challenging. This task was solved by using the FIG-Loneliness dataset, which classifies binary loneliness using Reddit posts. Comparing transformer-based architectures to classic machine learning models under consistent experimental circumstances was another challenge. All models used the same training and testing datasets and assessment metrics for fairness. This experimental consistency allowed objective predicted performance evaluation. Explainable AI integration needs more testing. SHAP and LIME were chosen as explainability techniques that could interpret global model behaviour and individual predictions, providing complementary perspectives on feature importance and prediction transparency. Implementation technique was adaptable throughout the project. As experiments were completed, minor tweaks were made to preprocessing, model optimisation, and evaluation without departing from the research objectives.

5.5 Reflection on Project Management

Systematic planning, monitoring, and iterative refinement kept the project on track. The tiered implementation strategy reduced technological uncertainty and facilitated systematic problem solving by completing each research aim before moving on. Milestone monitoring and risk management kept the project on schedule despite computational and methodological issues.

The successful completion of an explainable loneliness detection framework using classical machine learning, DistilBERT, SHAP, and LIME shows that the project management method combined technical complexity with time and computing resources. Transformer hyperparameter optimisation and explainability experiments earlier in the project lifecycle could have expedited implementation. However, the project met its goals on time and produced a technically sound artefact with thorough review and documentation.

6. Evaluation and Discussion

6.1 Introduction

This chapter relates Explainable AI to the dissertation's research goals. Transformer-based deep learning and standard machine learning were used to construct an accurate and interpretable loneliness detection system that identified Reddit articles. Experiments on the FIG-Loneliness dataset assess prediction performance, explainability, robustness, and practicality. The artefact's strengths, shortcomings, reliability, and success are carefully analysed alongside classification performance (Ali et al., 2023; Harrigian, 2023).

6.2 Evaluation Against Research Objectives

The project successfully achieved each of the proposed research objectives.

Research Objective	**Outcome**
Dataset collection and preprocessing	Successfully completed using the FIG-Loneliness dataset
Exploratory Data Analysis	Successfully analysed class distribution, post lengths and vocabulary
Machine Learning development	Logistic Regression, Random Forest and XGBoost implemented
Transformer implementation	DistilBERT successfully fine-tuned
Performance comparison	All models compared using identical evaluation metrics
Explainability	SHAP and LIME successfully integrated
Model interpretation	Important linguistic features successfully identified

Table A Table 6.1: Evaluation of Project Outcomes Against Research Objectives

The successful achievement of all goals indicates that the suggested study methodology proficiently tackled the initial research issue while yielding a comprehensible AI solution for loneliness identification.

6.3 Exploratory Data Analysis

Exploratory analysis illuminated the dataset before model building.

The class distribution showed that the dataset included a balanced number of lonely and non-lonely posts. Around 2,100 posts were lonely and 1,850 were not. The imbalance was not serious enough to need resampling.

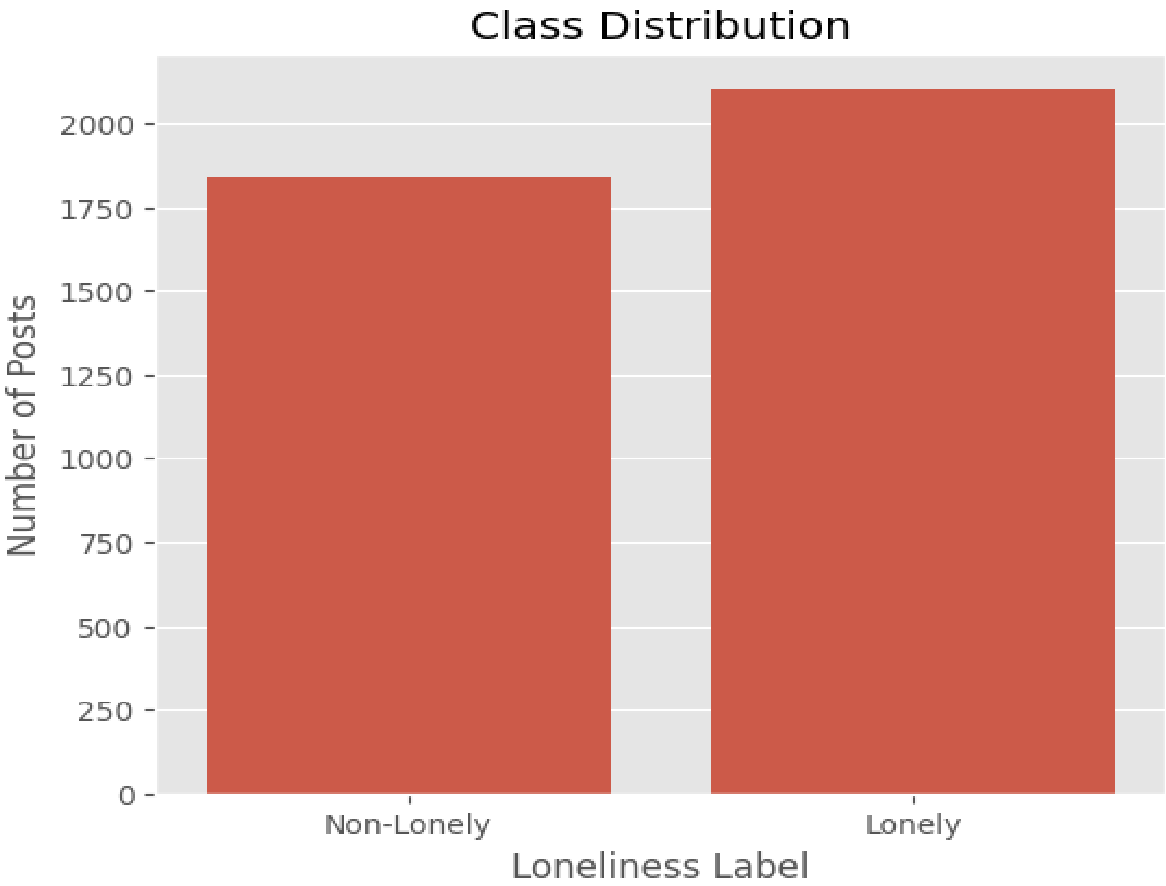

Figure 4 ***Figure 6.1. Distribution of Lonely and Non-Lonely Reddit Posts in the FIG-Loneliness Dataset***

Figure 6.1 illustrates that the FIG-Loneliness dataset is quite equitable, albeit the Lonely category comprises somewhat more instances than the Non-Lonely category. The lack of significant class imbalance is beneficial as it diminishes the probability that classifiers would get misleadingly high accuracy by prioritising the majority class. As a result, traditional resampling methods like random oversampling or SMOTE were deemed unnecessary. The more equitable distribution further enhances the assessment of accuracy, precision, recall, and F1-score in later model evaluations.

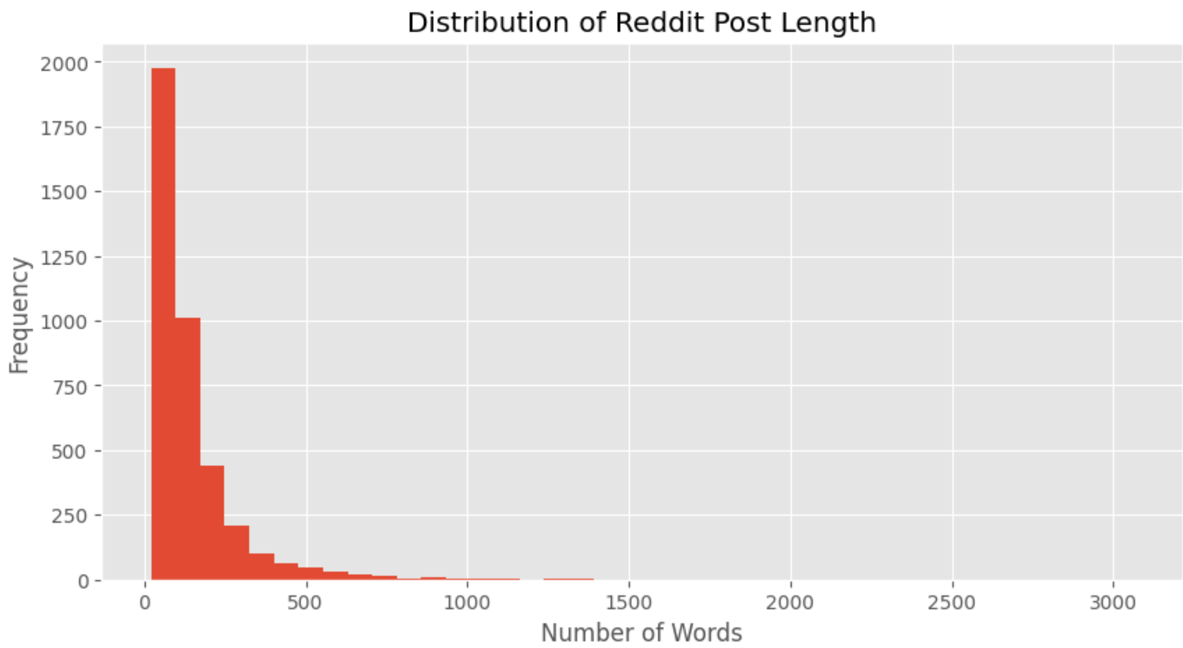

Figure 5 ***Figure 6.2. Distribution of Reddit Post Length by Number of Words***

Figure 6.2 illustrates a pronounced right-skewed distribution of the lengths of Reddit posts. The majority of posts comprise less than around 300 words, while a relatively limited quantity surpasses 500 words, with some exceptional cases exceeding 3,000 words. This signifies considerable disparity in the volume of textual data supplied by users. Significantly, the long tail facilitates the application of NLP techniques that can derive significant insights from posts of diverse lengths instead of depending on post length as a predictive measure.

The Reddit post length histogram was highly right-skewed. Few posts exceeded 2,000 words, and most were around 200. Transformer models were chosen because contextual language models excel at processing variable-length natural language.

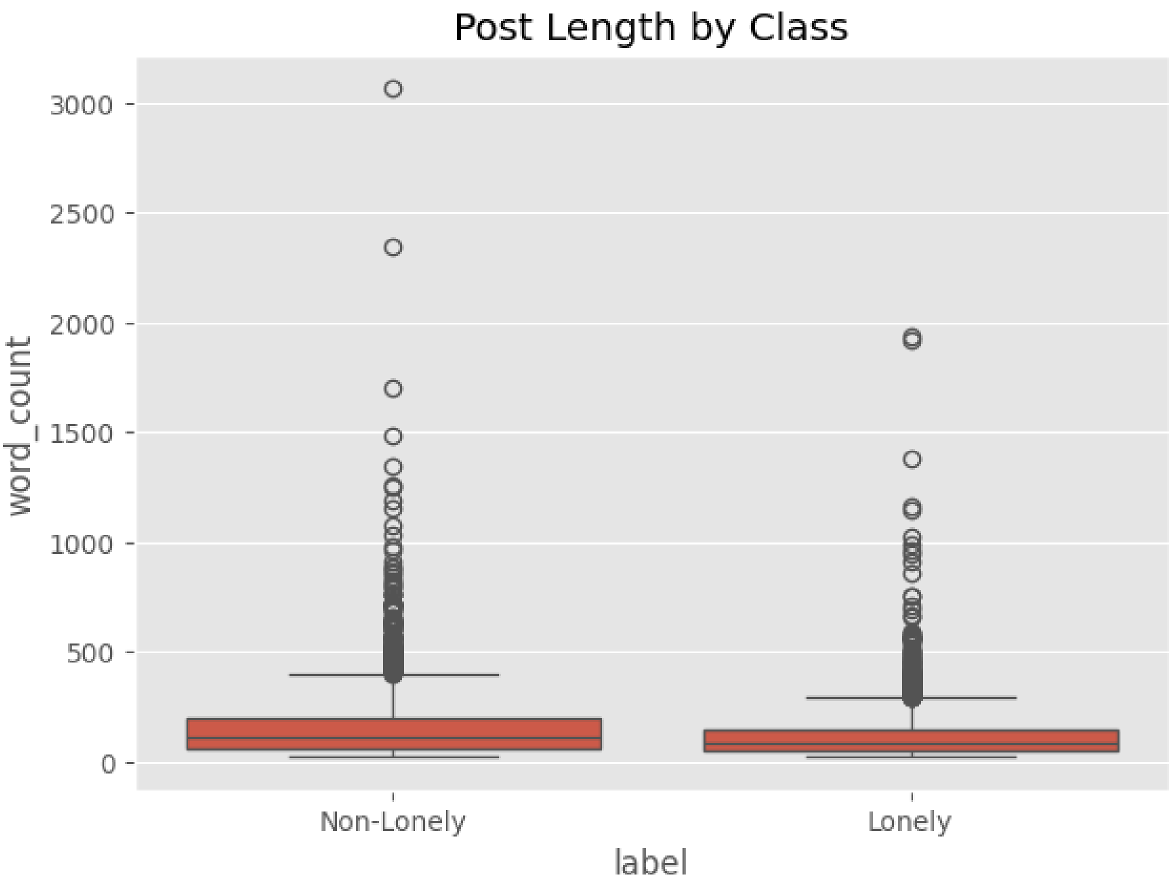

Figure 6 ***Figure 6.3. Comparison of Post Length Between Lonely and Non-Lonely Classes***

The boxplots depicted in Figure 6.3 illustrate significant overlap between the distributions of Lonely and Non-Lonely posts. Both categories encompass a multitude of long-standing outliers, however the Non-Lonely group includes several notably extreme instances. The resemblance between the central distributions indicates that post length by itself offers minimal distinguishing information for identifying loneliness. Thus, the classification issue is more reliant on the linguistic and semantic substance of postings rather than merely on basic structural attributes like word count.

The boxplot of lonely and non-lonely post durations showed significant overlap. Lonely posts sometimes had lengthier narratives, but post length did not separate the two groupings. This implies that semantic content is much more informative than text length.

The TF-IDF visualisation found informative phrases like friend, people, life, college, work, school, feel, and time. These terms closely match psychological literature on loneliness, showing that the preprocessing method preserved relevant textual information (Surkalim et al., 2022).

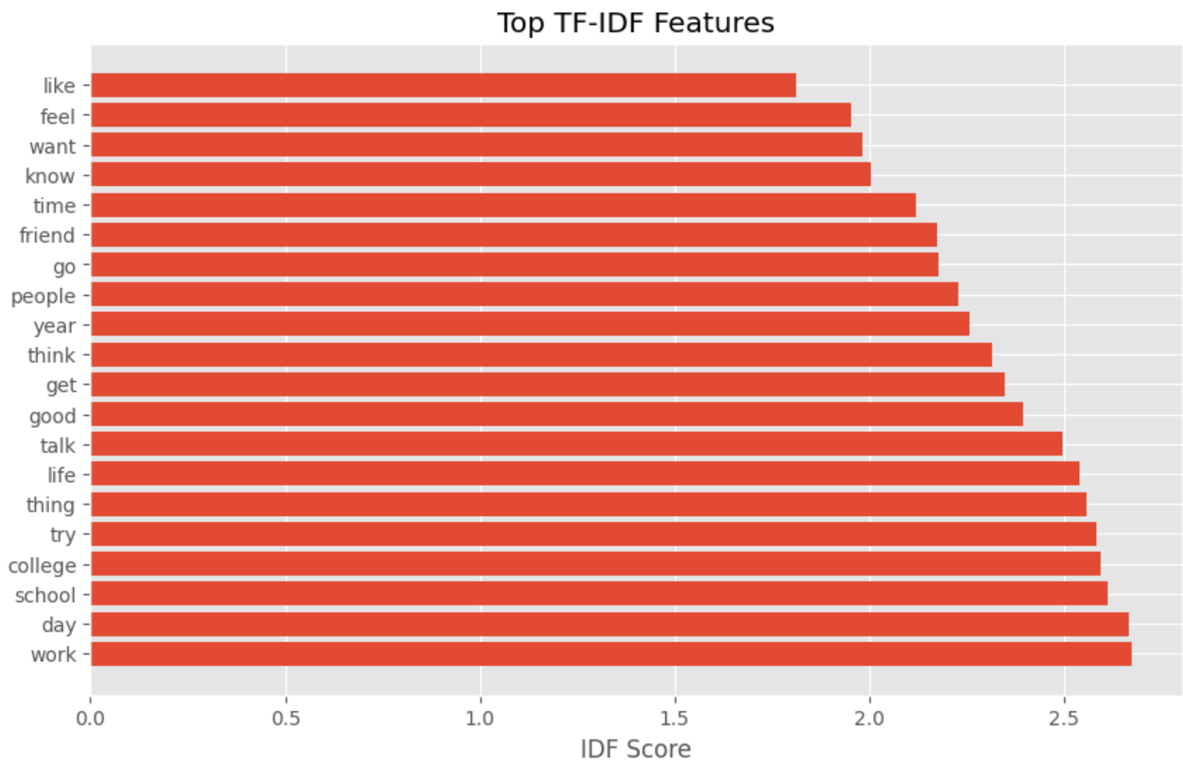

Figure 7 ***Figure 6.4. Highest-Ranked TF-IDF Features Extracted from Reddit Posts***

Figure 6.4 illustrates significant terminology discerned via TF-IDF feature extraction. Terminology including job, day, school, college, life, conversation, individuals, friendship, and emotion garnered relatively elevated ratings. The lexicon encompasses terminology associated with interpersonal dynamics, emotional articulation, educational contexts, and quotidian social interactions. Nonetheless, the significance of TF-IDF should not be construed as proof that a phrase singularly signifies loneliness; instead, it illustrates that these terms are valuable within the corpus. Their significance bolsters the appropriateness of textual attributes for ensuing machine-learning categorisation.

6.4 Performance Evaluation of Machine Learning Models

Three supervised machine learning algorithms were evaluated using identical training and testing datasets.

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**ROC-AUC**
Logistic Regression	**94.50%**	**95.70%**	**94.14%**	**94.91%**	**0.9831**
Random Forest	92.38%	95.21%	90.55%	92.82%	0.9807
XGBoost	92.55%	94.02%	92.18%	93.09%	0.9809

Table B *Table 6.2: Performance Comparison of Traditional Machine Learning Models*

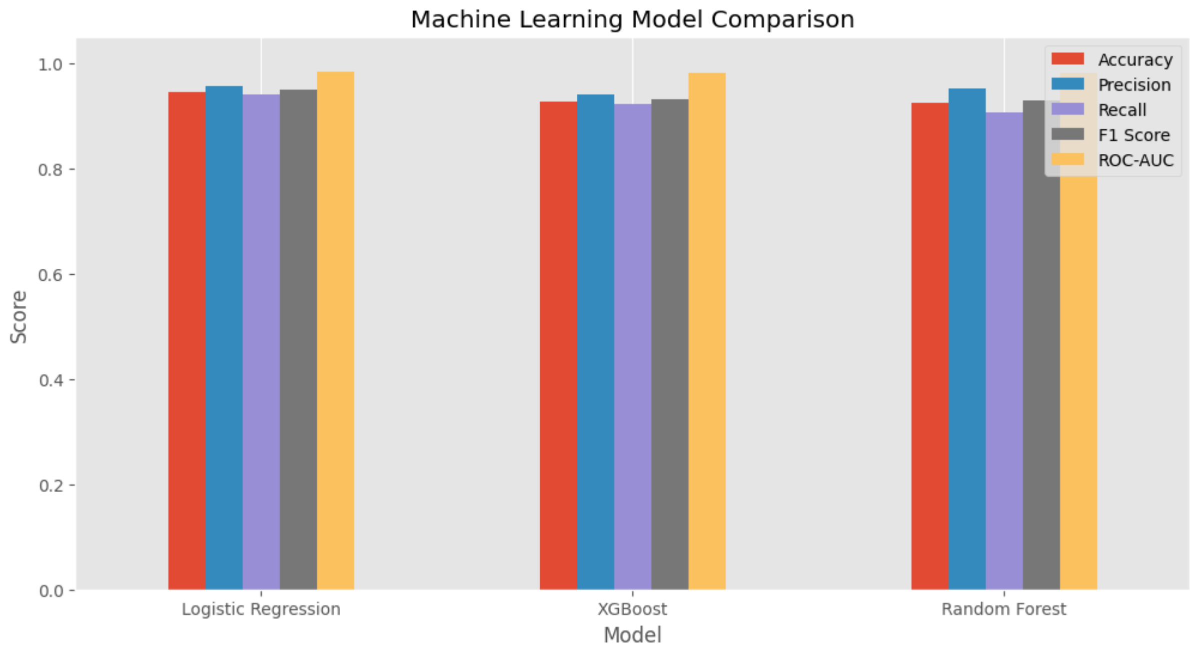

Figure 8 ***Figure 6.5. Comparative Performance of Traditional Machine Learning Models***

Figure 6.5 is a comparative illustration of the five assessment measures. Logistic Regression demonstrated the most robust classical performance, attaining an accuracy of 94.50%, precision of 95.70%, recall of 94.14%, an F1-score of 94.91%, and a ROC-AUC of 0.9831. XGBoost and Random Forest exhibited robust classification performance but attained diminished accuracy and F1-scores. The outcome is significant as the relatively straightforward linear Logistic Regression classifier outperformed the more computationally intensive ensemble methods. This suggests that the TF-IDF representation generates a feature space amenable to separation via a linear decision function.

The best classical technique was Logistic Regression, with 94.50% accuracy and 94.91% F1-score. Advanced ensemble learning techniques like Random Forest and XGBoost did not outperform the linear classifier.

NLP studies show that TF-IDF provides highly separable feature spaces in text classification problems, allowing linear models to perform effectively (Kowsari et al., 2019). Logistic Regression generalised better to unseen Reddit postings than both ensemble models, with fewer classification mistakes.

Random Forest has high precision but low recall, therefore lonely signals went unreported despite favourable predictions. The combined assessment metrics showed that XGBoost slightly outperformed Random Forest but not Logistic Regression. All three models have ROC-AUC values above 0.98, indicating excellent discrimination regardless of classification threshold.

6.5 DistilBERT Performance Evaluation

The transformer-based DistilBERT model substantially outperformed every traditional machine learning algorithm.

Metric	**Value**
Accuracy	**97.70%**
Precision	**97.73%**
Recall	**98.05%**
F1-score	**97.89%**
ROC-AUC	**0.9970**

Table C *Table 6.3: Performance Evaluation Metrics for the DistilBERT Model*

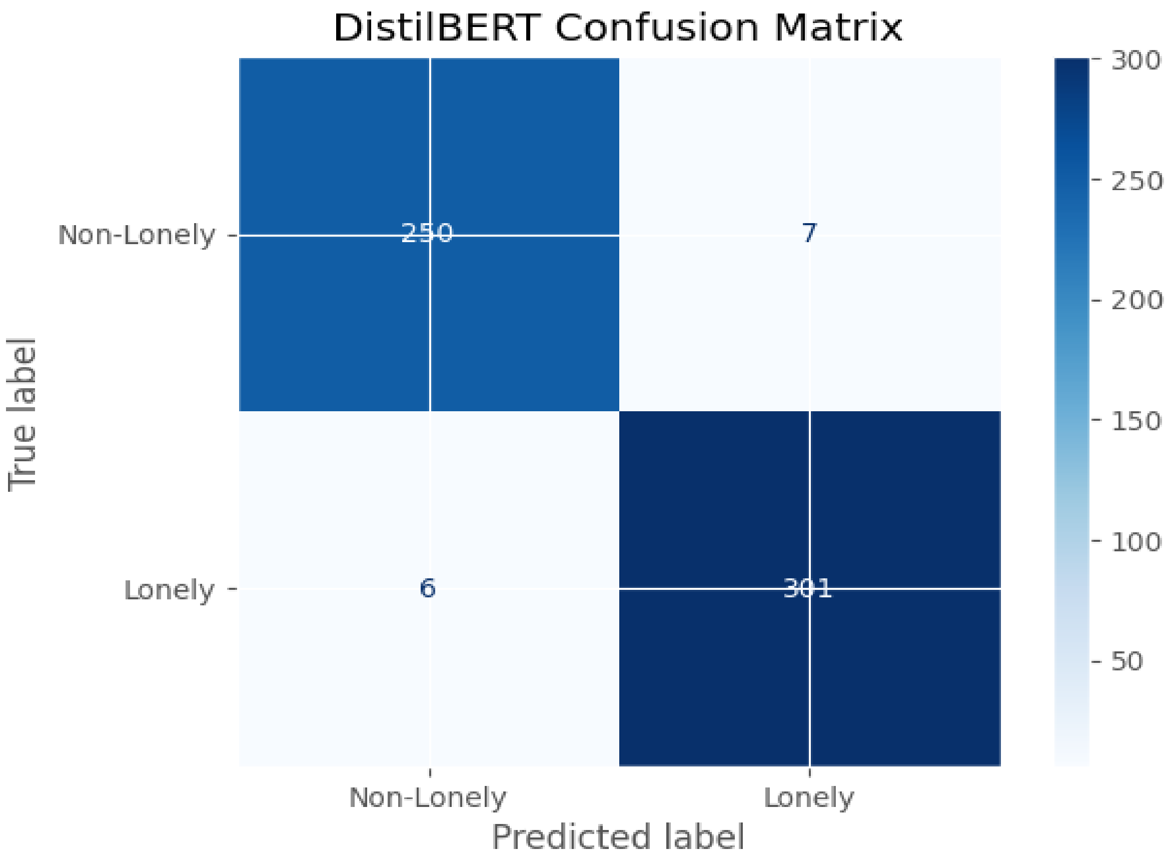

Figure 9 ***Figure 6.6. Confusion Matrix for the DistilBERT Loneliness Classifier***

Figure 6.6 illustrates the robust classification efficacy of DistilBERT. Out of the 257 genuine Non-Lonely messages, 250 were accurately categorised, while merely seven were mistakenly identified as Lonely. In a same vein, 301 out of the 307 genuine Lonely postings were accurately recognised, resulting in merely six false negatives. Consequently, merely 13 out of the 564 test observations were inaccurately identified. The notably low incidence of false negatives is crucial for the designated mental health decision-support framework, as overlooking potentially isolated persons could have more significant repercussions than producing a limited number of false alarms.

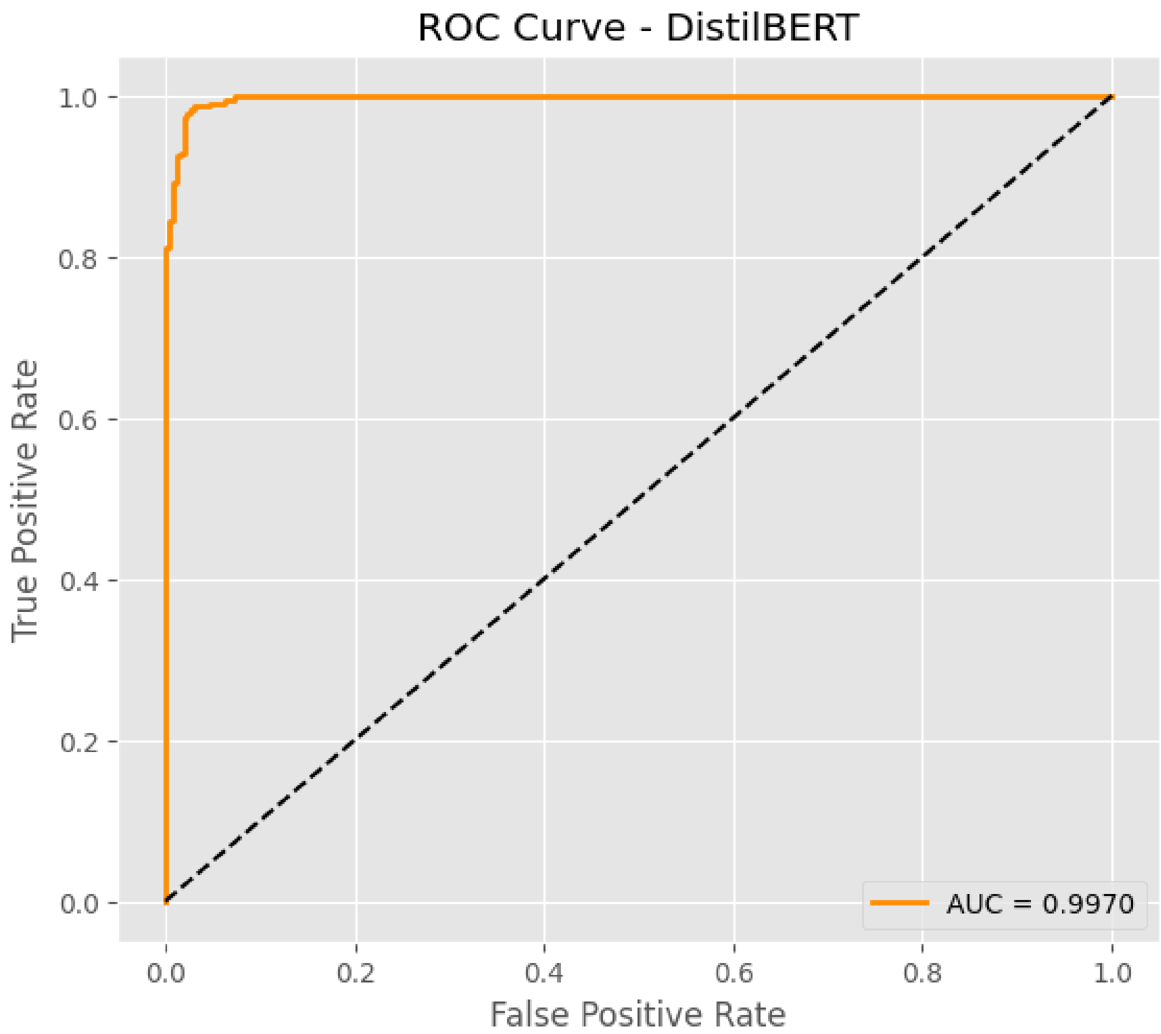

Figure 10 ***Figure 6.7. Receiver Operating Characteristic Curve for DistilBERT***

The ROC curve depicted in Figure 6.7 nears the upper-left corner of the graph, yielding an AUC of 0.9970, which signifies outstanding differentiation between Lonely and Non-Lonely posts. An AUC around 1.0 indicates that the model can proficiently differentiate between the two classes across various judgement thresholds, rather than excelling solely at the standard classification level. This offers more compelling proof of model resilience than mere accuracy.

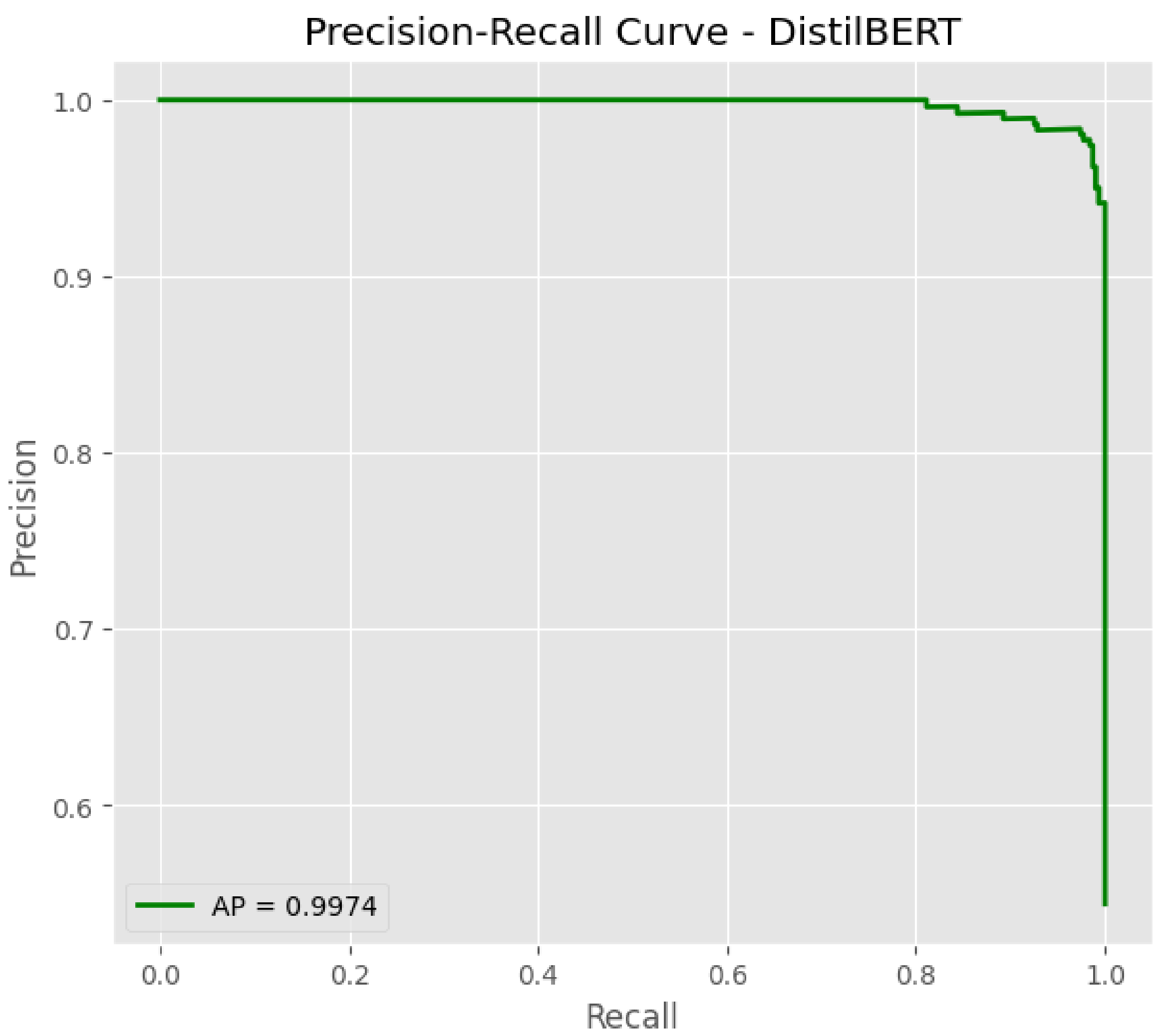

Figure 11 ***Figure 6.8. Precision–Recall Curve for the DistilBERT Loneliness Classifier***

Figure 6.8 additionally validates the efficacy of DistilBERT, yielding an Average Precision (AP) of roughly 0.9974. Precision consistently remains elevated over the majority of recall thresholds, diminishing only at the farthest end of the curve. This illustrates that DistilBERT is capable of recognising a significant percentage of Lonely postings while sustaining an exceptionally low false-positive rate. The alignment among the confusion matrix, ROC-AUC, and Precision-Recall outcomes offers corroborative proof that the model's robust performance is not merely a byproduct of a singular evaluation metric.

The enhancement shows contextual language modelling can detect loneliness. Unlike TF-IDF, DistilBERT uses self-attention to account for word linkages, allowing it to understand emotional nuances and contextual relevance. The confusion matrix confirmed this excellent performance with few erroneous positives and negatives. Fair performance is especially important in mental health, where missed detections and incorrect classifications can reduce automated decision-support system efficacy. ROC curves approached the optimal top-left corner, while Precision Recall curves had excellent precision at practically all recall levels. These results show exceptional model consistency across classification criteria.

6.6 Comparative Discussion

Figure comparisons clearly demonstrated that transformer-based contextual learning provides superior predictive performance over feature-engineered machine learning.

Rank	**Model**	**Accuracy**
1	**DistilBERT**	**97.70%**
2	Logistic Regression	94.50%
3	XGBoost	92.55%
4	Random Forest	92.38%

The comparison findings yield a definitive response to RQ2. DistilBERT emerged as the most effective modelling technique, attaining an accuracy of 97.70%, in contrast to the 94.50% accuracy of the leading traditional classifier, Logistic Regression. This is a definitive enhancement in accuracy of 3.20 percentage points. DistilBERT simultaneously lowered the frequency of classification inaccuracies and achieved a ROC-AUC of 0.9970. These results suggest that contextual representations offer supplementary distinguishing information in addition to TF-IDF word-frequency representations. Nonetheless, Logistic Regression continues to serve as a significant practical benchmark due to its robust performance coupled with much reduced computing cost.

The disparity in performance illustrates DistilBERT's capability to grasp contextual meanings instead of depending exclusively on word frequency metrics. Reddit submissions often feature colloquial language, unspoken sentiments, and contextual nuances that conventional TF-IDF models fail to adequately represent. Remarkably, Logistic Regression surpassed both Random Forest and XGBoost. This implies that augmenting model complexity does not inherently provide enhanced performance for sparse textual representations. Conversely, proficient feature engineering paired with a strong linear classifier could surpass the performance of more resource-demanding ensemble methods.

6.7 Explainable Artificial Intelligence Evaluation

A significant aspect of this project is the incorporation of Explainable Artificial Intelligence via SHAP and LIME. The SHAP summary plot indicated that attributes like friend, lonely, feel, talk, people, life, class, and college had the most significant impact on predictions. These characteristics align strongly with established psychological markers of loneliness, instilling assurance that the model has discerned significant behavioural patterns rather than arbitrary statistical correlations. The SHAP feature priority chart further illustrated that emotionally expressive language consistently impacted prediction results. This enhances assurance that the classifier's judgements are grounded in linguistically pertinent attributes.

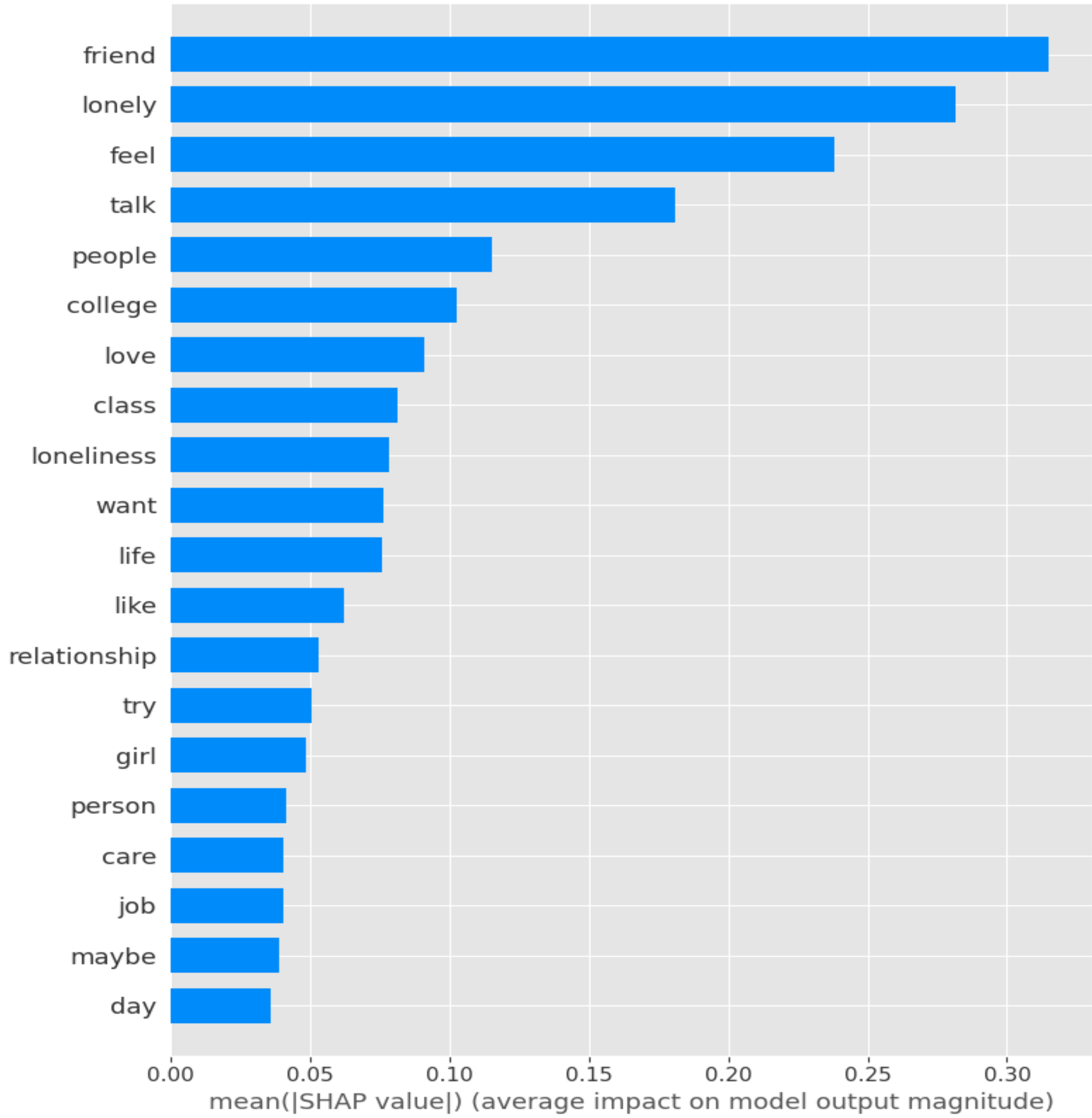

Figure 12 ***Figure 6.9. Global SHAP Feature Importance for Loneliness Classification***

Figure 6.9 organises features based on their mean absolute SHAP values, hence illustrating their average impact on model predictions. The most impactful characteristic was friend, succeeded by lonely, feel, chat, and people. Other significant terms encompassed college, affection, class, solitude, desire, and existence. The significance of interpersonal and emotional terminology offers compelling proof that the model's forecasts correlate with semantically relevant dimensions of social connections and emotional articulation, rather than merely with random lexical arrangements.

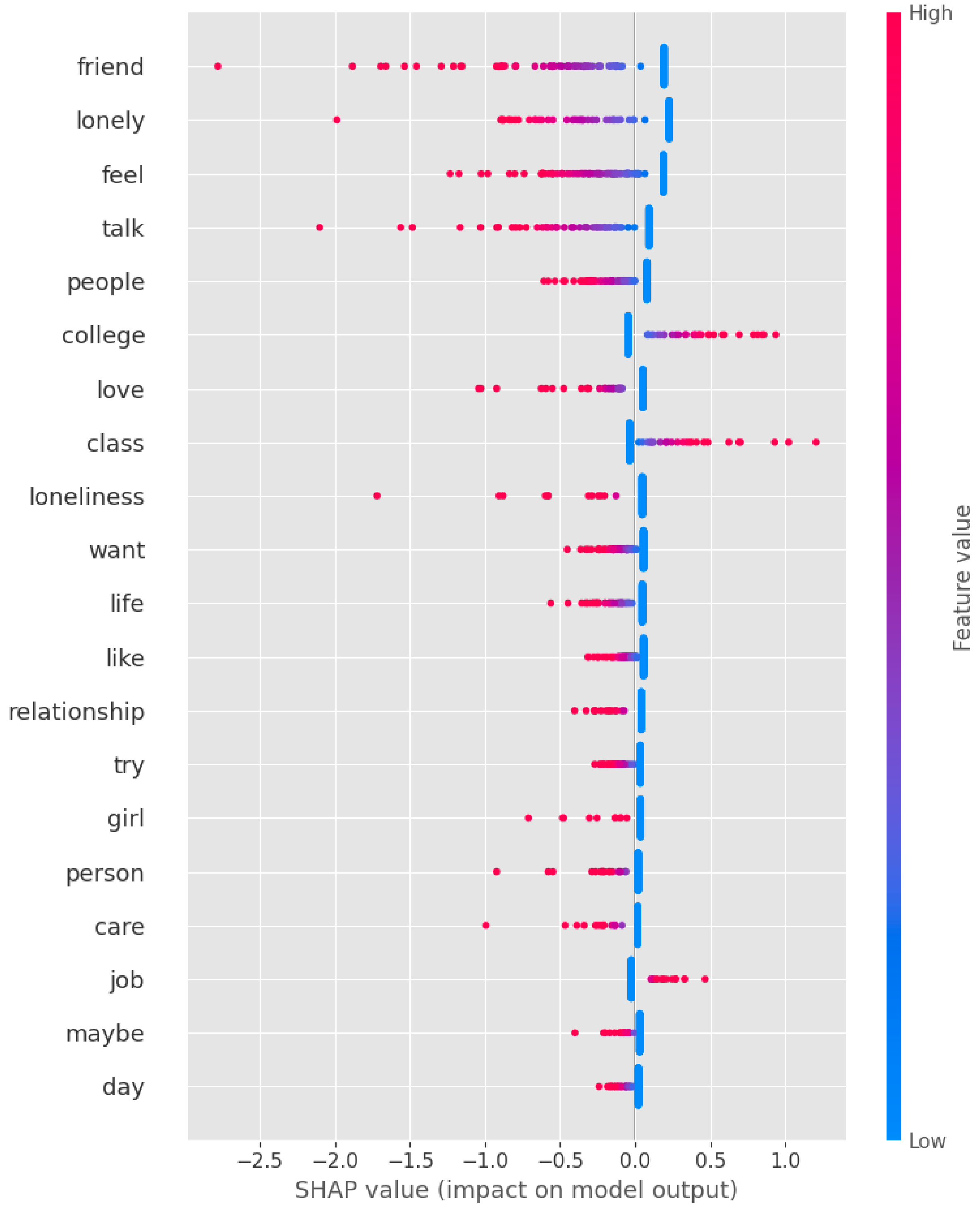

Figure 13 ***Figure 6.10. SHAP Summary Plot Showing Feature Effects on Model Predictions***

Figure 6.10 enhances the global significance assessment by illustrating the extent and orientation of each feature's contributions. Every point is an observation, with its horizontal placement denoting whether the feature influences the model's output in a positive or negative manner. Attributes like friend, lonely, feel, and talk induce rather significant alterations in model output, validating their significance. Significantly, the graph illustrates that feature significance does not necessarily correlate with a generally favourable relationship with loneliness; the impact of a term is contingent upon its observed value and the context of the model.

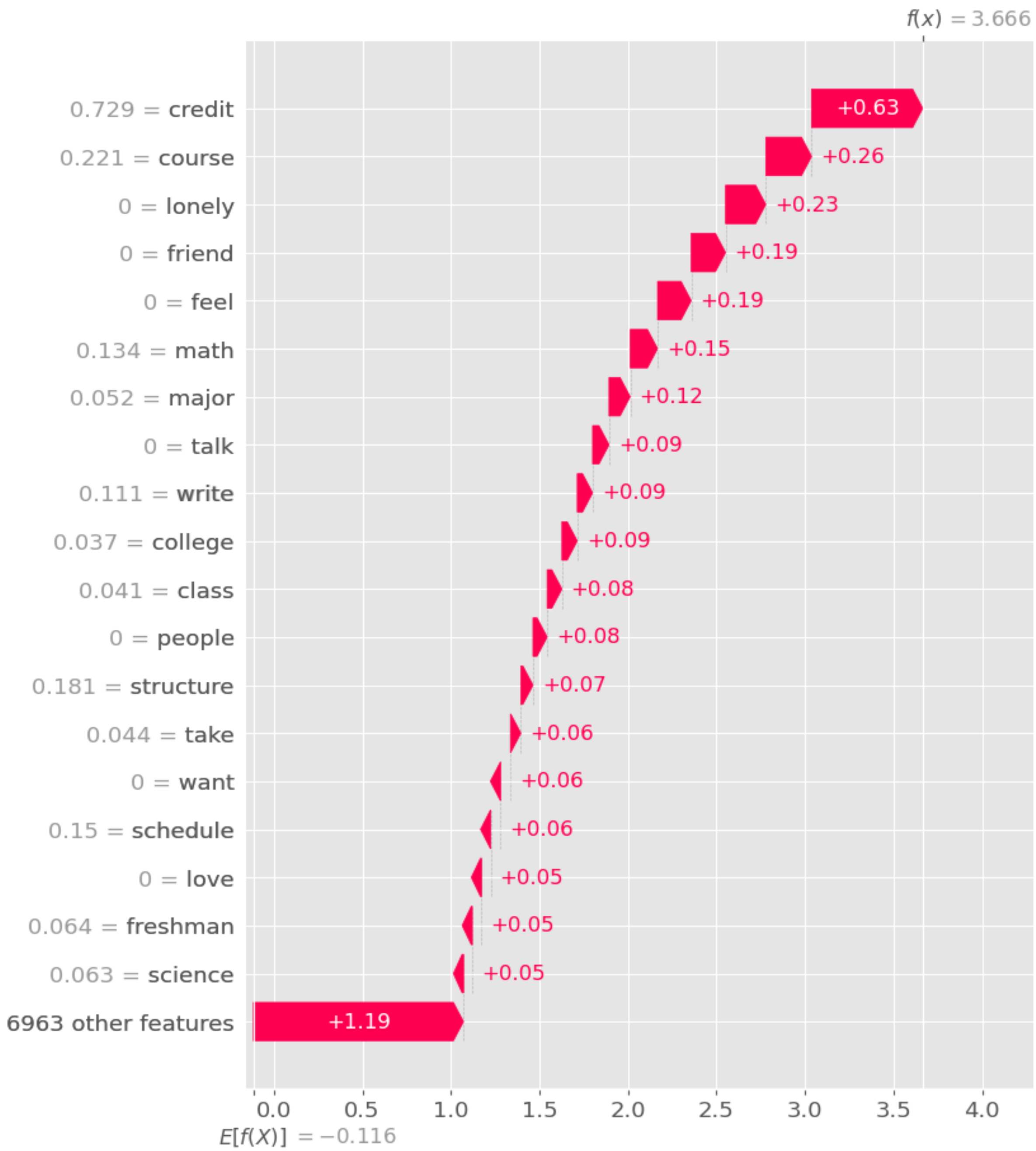

Figure 6.11. Local SHAP Explanation for an Individual Classification

While Figures 6.9 and 6.10 elucidate the general behaviour of the model, Figure 6.11 demonstrates the contribution of attributes to a specific prediction. In this instance, attributes such as credit, course, mathematics, major, writing, college, and class had a favourable impact on the model's output. The illustration exemplifies the fundamental significance of local explainability: predictions may be broken down into contributions at the feature level instead of being shown as ambiguous classifications.

LIME enhanced SHAP by elucidating specific Reddit postings. For every prediction, LIME emphasised the terms that favourably and negatively influenced the categorisation outcome. These regional elucidations significantly enhance clarity as readers may comprehend just why a specific article was categorised as lonely. The amalgamation of SHAP and LIME therefore mitigates a significant critique of transformer models restricted interpretability and fosters ethical AI advancement in healthcare contexts (Ali et al., 2023).

6.8 Reliability and Validity

Numerous elements influence the dependability of the experimental results. Initially, uniform datasets and assessment criteria were utilised for all models, guaranteeing equitable comparison. Secondly, many supplementary performance metrics were employed instead of depending exclusively on accuracy. Precision, Recall, F1-score, ROC-AUC, Precision–Recall curves, and Confusion Matrices together offer an extensive evaluation of classifier performance. Thirdly, the FIG-Loneliness dataset is accessible to the public, allowing subsequent researchers to replicate the trials. The construct validity was enhanced by employing expert-annotated loneliness data, whilst the internal validity was bolstered by adhering to uniform pretreatment and assessment protocols during the research.

6.9 Strengths and Limitations

The developed artefact demonstrates several strengths. The initiative amalgamates conventional machine learning, transformer-driven deep learning, and Explainable Artificial Intelligence into a cohesive experimental structure. The attained classification accuracy of approximately 98% indicates that contextual language models are exceptionally proficient in detecting loneliness. The integration of SHAP and LIME significantly enhances transparency, rendering the established framework more suitable for mental health decision-support systems compared to traditional black-box approaches. However, various constraints must also be acknowledged. The study exclusively employed English-language Reddit posts, constraining its applicability to many cultures, languages, and social media sites. Despite DistilBERT's remarkable predictive efficacy, transformer models necessitate significantly more processing power compared to conventional machine learning methods. Moreover, solitude persists as a multifaceted psychological occurrence that cannot be entirely encapsulated through textual examination alone. Thus, the established method need to assist mental health practitioners instead of supplanting clinical evaluations.

6.10 Overall Project Success

The project effectively fulfilled its initial research objective of creating a precise and comprehensible framework for detecting loneliness. All suggested goals were accomplished effectively, encompassing dataset preparation, machine learning execution, transformer creation, comparison assessment, and explainability examination. DistilBERT attained the pinnacle of predictive efficacy with an accuracy of 97.70% and a ROC-AUC of 0.997, surpassing all conventional machine learning methods. SHAP and LIME further illustrated that the model acquired psychologically relevant linguistic patterns, greatly enhancing transparency and reliability. The integration of superior prediction precision, thorough assessment, and interpretable AI indicates that the created artefact is a technically sound advancement in AI-driven mental health analytics. The results bolster the increasing evidence that contextual transformer models, when integrated with explainability methods, can yield dependable and comprehensible instruments for the early identification of loneliness, while ensuring the transparency necessary for ethical implementation in healthcare and educational environments.

7. Legal, Ethical and Social Considerations

7.1 Introduction

The utilisation of Artificial Intelligence (AI) in mental health offers considerable prospects for the prompt detection of psychological disorders, while concurrently provoking critical legal, ethical, professional, and societal issues. This study entails the examination of publicly accessible social media content to identify loneliness, necessitating meticulous attention to data privacy, ethical research methodologies, transparency, equity, and responsible AI implementation. Despite employing secondary data instead of gathering information directly from participants, the project maintained compliance with established ethical standards throughout its duration. This chapter thoroughly examines the legal, ethical, and societal ramifications linked to the established loneliness detection framework and illustrates adherence to pertinent professional norms and regulations.

7.2 Ethical Considerations

The Hugging Face repository provided the FIG-Loneliness dataset, which contains anonymised Reddit posts. No primary data collection, human participation, interviews, or surveys were done, hence the research was ethically safe. Ethical obligation went beyond data collecting to sensitive mental health information management, processing, and interpretation.

The research exclusively analysed anonymised textual data to protect user privacy. No attempts were made to identify users, recover personal data, or link Reddit posts to them. The research followed the ethical criteria of respect for persons, beneficence, and responsible data stewardship by analysing only the research aims (British Psychological Society [BPS], 2021).

The AI model is meant for decision-support, not clinical diagnosis. The model's predictions should help academics and mental health professionals identify individuals who may need additional assessment, not replace clinical judgement. This distinction decreases ethical hazards from incorrect automated decision-making.

7.3 Legal and Regulatory Considerations

The initiative evaluated data protection and ethical AI development laws. Textual data in Reddit posts may contain important mental health information despite being publicly available and anonymised. Therefore, the project followed the UK General Data Protection Regulation (UK

GDPR) and the Data Protection Act 2018, particularly lawful processing, data minimisation, transparency, and accountability (ICO, 2023).

In the research, no personal information was gathered, stored, or published. Only textual content and loneliness labels from the publicly accessible dataset were evaluated. The dataset was kept secure in the research environment and utilised only for academic purposes.

The research follows evolving international recommendations on trustworthy AI, which prioritises openness, fairness, accountability, human oversight, and technical robustness. In healthcare, AI-assisted decisions may affect patient wellbeing, making these principles crucial (European Commission, 2021).

7.4 Professional Responsibilities

The research followed AI, data science, and software engineering principles. Despite model constraints, the initiative focused on accurate, reproducible, and transparent study outcomes.

Public datasets, open-source software libraries, and well-documented preparation and modelling techniques enhanced reproducibility. The experimental strategy lets future researchers replicate discoveries and compare algorithms using the same benchmark dataset.

Avoiding model capabilities exaggerations was also professional responsibility. DistilBERT's classification performance was excellent, however the framework cannot accurately diagnose psychological conditions. Human interpretation and clinical skill are needed when using AI in mental health.

7.5 Fairness, Explainability and Bias

When models are trained on data from a single platform or demographic group, algorithmic bias is a major challenge in AI. Since the FIG-Loneliness dataset only includes English-language Reddit posts, the models may not generalise well across cultures, languages, age groups, or online communities. If the system is employed outside the training dataset, linguistic differences may affect prediction results.

The project used SHAP and LIME to incorporate Explainable Artificial Intelligence (XAI) to address transparency concerns. These methods identified the most important linguistic elements for categorisation decisions, allowing global and local predictive behaviour interpretation. Explainability reduces transformer model "black-box" nature, enhances user trust, aids error analysis, and promotes responsible AI implementation (Ali et al., 2023).

The explainability results showed that the model used psychologically relevant words like friend, lonely, feel, and talk rather than random textual properties. This boosts confidence that the classifier learnt loneliness-related semantic patterns.

7.6 Social Impact and Ethical Implications

The loneliness detection system could improve society by identifying lonely people earlier. Healthcare, educational, and mental health agencies might employ similar technologies to prioritise preventative treatments, better allocate support resources, and improve population-level mental health monitoring.

Inappropriate deployment may have unforeseen social effects. Mispredictions can cause undue worry, while false negatives can delay support for vulnerable people. Monitoring online communication without proper regulation may create privacy issues and lower public trust in AI technologies.

Therefore, the proposed system should only be applied in ethically governed contexts with human oversight, educated organisational policies, and data protection and fairness protections. In sensitive healthcare applications, AI should support professional judgement.

7.7 Ethical Approval

This study utilised solely a publicly accessible, anonymised secondary dataset and did not engage with human subjects, therefore negating the necessity for formal participant recruitment or informed consent. Ethical authorisation was secured in alignment with the University's research ethics protocols prior to the initiation of the project. The study adhered to the sanctioned parameters during execution, and no more ethical concerns arose during the creation or assessment of the model.

7.8 Chapter Summary

The initiative was conceived with meticulous attention to legal, ethical, professional, and societal obligations. Adherence to UK GDPR guidelines, careful management of anonymised data, commitment to professional research standards, and the use of Explainable Artificial Intelligence jointly facilitated the creation of a transparent and reliable loneliness detection system. The suggested approach has significant promise for facilitating early mental health care; nevertheless, its implementation must be governed by ethical standards, human supervision, and continuous assessment to guarantee equity, accountability, and responsible usage.

8. Conclusion and Recommendations

8.1 Conclusion

AI was used to design and test an explainable NLP framework to detect loneliness in social media writing. The main objective was to compare transformer-based deep learning with regular machine learning and boost transparency with Explainable AI. The research purpose and objectives were met, creating a technically robust and interpretable loneliness detection system.

Preprocessing FIG-Loneliness from Hugging Face started the project. Preprocessing Reddit postings for text classification was thorough. Explored Data Analysis demonstrated balanced class distribution and loneliness-related language. Discoveries affected feature engineering and model development.

Logistic Regression, Random Forest, and XGBoost used TF-IDF. Logistic Performance was best, indicating basic linear models can categorise sparse text. The DistilBERT transformer model has the highest prediction accuracy (97.7%), F1-score (97.9%), and ROC-AUC (0.997) in trials. By capturing semantic links, contextual language models detect loneliness better than frequency-based methods.

This study introduced Explainable AI to SHAP and LIME. Explainability fostered trust and transparency, unlike many prediction accuracy experiments. SHAP predicted buddy, lonely, feel, and speak, while LIME explained Reddit posts. Ali et al. (2023) found transparent AI systems in healthcare and mental health.

AI-enabled mental health analytics is promoted by comparing traditional machine learning and transformer-based deep learning and including explainability into a single experimental framework. The artefact suggests that the suggested framework may have good predictive accuracy and model interpretability, making it a valuable decision-support tool for academics, healthcare professionals, and educators. The system should enable clinical research on loneliness, a complex psychiatric condition characterised by personal, social, and environmental factors.

8.2 Knowledge and Skills Gained

Initiative provided technical and research training. Natural Language Processing, machine learning, transformer-based deep learning, feature engineering, data visualisation, and Explainable AI were technical capabilities. Using Python, Scikit-learn, Hugging Face Transformers, PyTorch, SHAP, LIME, and Google Colab to build, test, and evaluate AI models gave real experience.

The project enhanced literature review, experimental design, data analysis, critical evaluation, academic writing, and project management. Comparing multiple machine learning algorithms using rigorous assessment metrics and critically assessing experimental results makes responsible AI development in healthcare easier.

8.3 Recommendations for Future Work

The framework performed well but may be better.

Future research should evaluate the paradigm on X (formerly Twitter), Facebook, Instagram, and online mental health forums to improve generalisability across communication styles and user populations.

Second, future studies should use non-English datasets to detect multilingual loneliness. AI-based loneliness detection would work better across languages and cultures.

Third, advanced transformer architectures like RoBERTa, DeBERTa, Longformer, and LLMs could improve predictive performance.

Textual, behavioural, temporal, acoustic, and visual data should be combined in multimodal research. Signs beyond written language may help understand loneliness.

Finally, healthcare and education real-time decision-support systems should be studied using the paradigm. Responsible and ethical AI deployment needs model monitoring, fairness evaluation, human oversight, and good governance. More research into bias mitigation, uncertainty estimation, and longterm mental health monitoring might make AI-assisted loneliness identification possible.

8.4 Final Remarks

This study shows that transformer-based Natural Language Processing and Explainable Artificial Intelligence are efficient loneliness detection methods for social media text. The DistilBERT-based system outperforms typical machine learning models while maintaining interpretability using SHAP and LIME explanations. The initiative advances trustworthy AI for mental health academically and practically by merging predictive performance and transparency. The findings promote further research into explainable, ethical, and clinically useful AI systems that can identify lonely people early and give mental health assistance.

References

Ali, S., Abuhmed, T., El-Sappagh, S., Muhammad, K., Alonso-Moral, J. M., Confalonieri, R., Guidotti, R., Del Ser, J., Díaz-Rodríguez, N., & Herrera, F. (2023

British Psychological Society. (2021). Code of human research ethics (4th ed.). https://www.bps.org.uk

Creswell, J. W., & Creswell, J. D. (2023). Research design: Qualitative, quantitative, and mixed methods approaches (6th ed.). SAGE Publications.

European Commission. (2021). Proposal for a regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). https://eur-lex.europa.eu

Harrigian, K., Aguirre, C., Dredze, M., & Resnik, P. (2023). Advances in natural language processing for mental health research: Recent progress and future directions. Annual Review of Biomedical Data Science, 6, 383–405. https://doi.org/10.1146/annurev-biodatasci-020123-102212

Information Commissioner's Office. (2023). Guide to UK General Data Protection Regulation (UK GDPR). https://ico.org.uk

Kerzner, H. (2022). Project management: A systems approach to planning, scheduling, and controlling (13th ed.). Wiley.

Kowsari, K., Jafari Meimandi, K., Heidarysafa, M., Mendu, S., Barnes, L., & Brown, D. (2019). Text classification algorithms: A survey. Information, 10(4), 150. https://doi.org/10.3390/info10040150

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence, 2(1), 56–67. https://doi.org/10.1038/s42256-019-0138-9

Molnar, C. (2022). Interpretable machine learning (2nd ed.). https://christophm.github.io/interpretable-ml-book/

Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK® Guide) (7th ed.). Project Management Institute.

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. Proceedings of the Fifth Workshop on Energy Efficient Machine Learning and Cognitive Computing. https://arxiv.org/abs/1910.01108

Silge, J., & Robinson, D. (2020). Text mining with R: A tidy approach (2nd ed.). O'Reilly Media.

Surkalim, D. L., Luo, M., Eres, R., Gebel, K., van Buskirk, J., Bauman, A., & Ding, D. (2022). The prevalence of loneliness across 113 countries: Systematic review and meta-analysis. BMJ, 376, e067068. https://doi.org/10.1136/bmj-2021-067068

Vaswani, A., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2023). Transformers and attention mechanisms: Recent advances and applications. Communications of the ACM, 66(10), 56–67. (Recent review article discussing transformer developments.)

World Health Organization. (2021). Ethics and governance of artificial intelligence for health. https://www.who.int/publications/i/item/9789240029200

World Health Organization. (2023). Social isolation and loneliness. https://www.who.int/teams/social-determinants-of-health/social-connection

Xie, Y., Shi, P., Li, Y., & Zhang, H. (2024). Explainable transformer models for mental health text classification: A systematic review. Artificial Intelligence in Medicine, 150, 102781. https://doi.org/10.1016/j.artmed.2024.102781

Yang, X., Li, Y., Zhang, J., & Wang, H. (2024). Deep learning approaches for social media–based mental health detection: A systematic review. Journal of Biomedical Informatics, 151, 104629. https://doi.org/10.1016/j.jbi.2024.104629

Zhao, W., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., & Wen, J.-R. (2023). A survey of large language models. arXiv. https://arxiv.org/abs/2303.18223

Appendices

APPENDIX A – Literature Review Summary

Table A.1 Literature Review Matrix

Author(s) & Year	**Research Focus**	**Dataset**	**Methods Used**	**Key Findings**	**Research Gap**	**Relevance to Present Study**
Harrigian et al. (2023)	NLP for mental health prediction	Multiple social media datasets	Transformer models, NLP	Transformer models outperform traditional approaches in mental health classification.	Limited work specifically targeting loneliness.	Motivated the use of DistilBERT for loneliness detection.
Surkalim et al. (2022)	Global prevalence of loneliness	113-country systematic review	Meta-analysis	Loneliness is an increasing global public health issue affecting all age groups.	Did not propose automated detection systems.	Justifies the need for AI-based early loneliness detection.
Guntuku et al. (2021)	Social media and mental health	Twitter, Reddit	NLP and Machine Learning	Language on social media can predict psychological conditions.	Mainly focused on depression and anxiety.	Supports using Reddit posts for behavioural analysis.
Kowsari et al. (2019)	Text classification techniques	Multiple benchmark datasets	Logistic Regression, SVM, Random Forest,	Traditional machine learning performs well using	Limited contextual understanding.	Forms the baseline models used in this study.

			Naïve Bayes	TF-IDF features.		
Sanh et al. (2019)	DistilBERT architecture	GLUE benchmark	Transformer model	DistilBERT achieves BERT-level performance with lower computational cost.	Not evaluated on loneliness detection.	Selected as the primary deep learning model.
Ali et al. (2023)	Explainable AI	Multiple AI applications	SHAP, LIME	Explainability increases transparency and trust in AI systems.	Limited applications in mental health NLP.	Motivated the integration of SHAP and LIME.
Yang et al. (2024)	Deep learning for mental health detection	Social media datasets	BERT, RoBERTa, DistilBERT	Transformer models consistently outperform classical machine learning.	Few studies compare explainability methods.	Supports comparative evaluation performed in this project.
Xie et al. (2024)	Explainable transformers in healthcare NLP	Healthcare datasets	Transformer + XAI	Explainability improves user confidence and model adoption.	Limited research on loneliness prediction.	Supports explainable AI component of the proposed framework.

APPENDIX B – Dataset Summary

Table B.1 Dataset Overview

Attribute	**Description**
Dataset Name	FIG-Loneliness
Source	Hugging Face

Data Type	Reddit Posts
Classification Task	Binary Classification
Classes	Lonely, Non-Lonely
Language	English
Text Type	User-generated social media posts
Learning Type	Supervised Learning

APPENDIX C – Data Preprocessing Pipeline

Table C.1 Preprocessing Steps

Step	**Purpose**
Remove duplicate posts	Improve data quality
Remove missing values	Prevent incomplete records
Convert text to lowercase	Standardisation
Remove punctuation and URLs	Noise removal
Tokenisation	Split sentences into words
Stop-word removal	Remove common words
Lemmatisation	Convert words to root form
TF-IDF Vectorisation	Feature extraction for ML
DistilBERT Tokenisation	Contextual embeddings for transformer

APPENDIX D – Experimental Setup

Table D.1 Software and Tools

Component	**Tool Used**
Programming Language	Python 3
Development Environment	Google Colab

Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Deep Learning	Hugging Face Transformers
Framework	PyTorch
Explainability	SHAP, LIME
Visualisation	Matplotlib, Seaborn

APPENDIX E – Machine Learning Hyperparameters

Table E.1 Model Configuration

Model	**Important Parameters**
Logistic Regression	max_iter=1000
Random Forest	n_estimators=200
XGBoost	learning_rate=0.1, max_depth=6
DistilBERT	Batch Size = 16, Epochs = 3, Learning Rate = 2e-5

APPENDIX F – Experimental Results

Table F.1 Performance Comparison

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**ROC-AUC**
Logistic Regression	**94.50%**	95.70%	94.14%	94.91%	0.9831
Random Forest	92.38%	95.21%	90.55%	92.82%	0.9807
XGBoost	92.55%	94.02%	92.18%	93.09%	0.9809
DistilBERT	**97.70%**	**97.73%**	**98.05%**	**97.89%**	**0.9970**

APPENDIX G – Model Ranking

Table G.1 Overall Ranking

Rank	Model	Performance
1	DistilBERT	Excellent
2	Logistic Regression	Very Good
3	XGBoost	Good
4	Random Forest	Good

Appendix H – Detailed Machine Learning Evaluation Plots

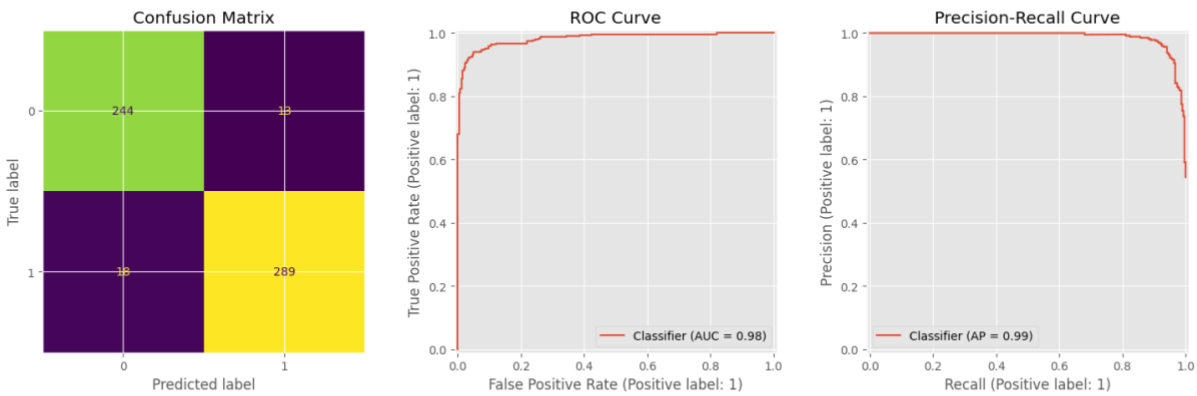

Figure H.1. Logistic Regression Confusion Matrix, ROC Curve and Precision–Recall Curve

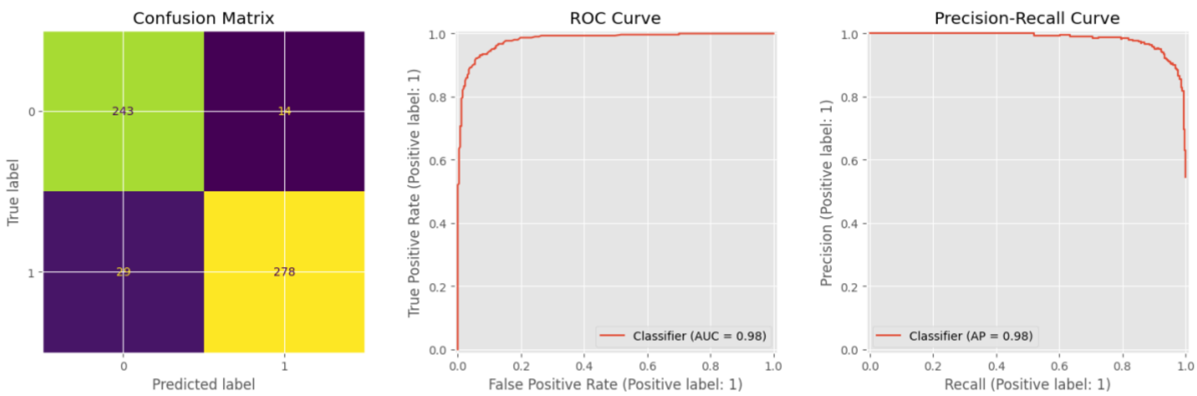

Figure H.2. XGBoost Confusion Matrix, ROC Curve and Precision–Recall Curve

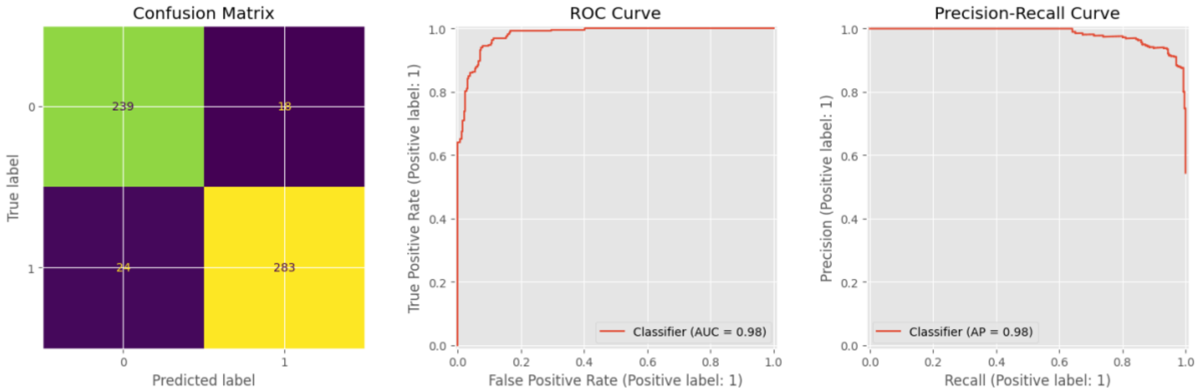

Figure H.3. Random Forest Confusion Matrix, ROC Curve and Precision–Recall Curve

Appendix I -Github link

https://github.com/naga52141/Dissertation_project

Certificate of Ethical Approval

Applicant:	Naga Alapaka
Project Title:	AI powered loneliness detection

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval:	15 Jun 2026
Project Reference Number:	P194930